

At Last, Empirical Elicitations of the Magnitudes of those Risks (and Costs!) too Small to Matter and those too Large to Abide

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ABSTRACT

Scholars, decision-makers, and pundits have long tossed around words and numbers meant to demarcate those health/safety risks that are too small to matter at all (called, variously, “trivial” or “*de minimis*” or “insignificant”), or risks that are so large as to be abhorrent (“unacceptable,” “intolerable,” and the like). But there has been nothing approaching a consensus about how large or small these important probabilities are, and worse, there have been no serious attempts to explore these concepts via any carefully-elicited preferences from laypeople themselves. Perhaps even more concerning, the dominant paradigm—cost-benefit analysis (CBA)—requires balancing of (monetized) risks against the costs of controlling them, but the parallel concepts of *de minimis costs* and *intolerable costs* have never been formally considered. As a result of these lapses, all attempts to consider equity or justice in the *distribution* of how risks and costs are borne are farcical, despite claims to the contrary.

This article reports on a pair of elicitations involving 1350 laypeople. In one elicitation, a careful set of background and contextual explanations and a structured series of questions probed where each subject perceived lifetime excess mortality risk (from involuntary exposure) to become “so small you would not spend a moment thinking about it,” and where it becomes so large that “you would strongly object, as an ethical matter, to ever having to face it.” With a separate sample, we asked the parallel questions with regard to regulatory costs imposed by the government in the name of risk reduction.

Although there was significant interindividual variation in responses, the central tendency of each of these four science-policy-values “goalposts” was as follows: (1) *de minimis* risk—a lifetime excess chance of premature death of 1 in 770,000; (2) intolerable risk—1 chance in 570; (3) *de minimis* cost—\$25 per year; and (4) intolerable cost—\$775 per year. These results offer, for the first time, a principled and evidence-based starting point for further refinement of which risks or costs to individuals can/should be disregarded in setting policy, and which must dominate policies regardless of how total “monetized lives saved” and total dollar costs compare to each other when viewed simplistically.

We offer several prescriptions for how CBA might change, and how environmental/health/safety policy might improve, to take account of these fundamental thresholds in how risks and costs actually impose burdens on citizens. Both those interventions that impose intolerable costs while only reducing diffuse and trivially small risks, *and* those failures to intervene (sufficiently) that perpetuate intolerable risks for the sake of avoiding trivial and eminently manageable costs, are inefficient, unjust, and are thus unethical.

Keywords: risk assessment, stated preference, cost-benefit analysis, *de minimis* risk, environmental equity

1. Introduction:

We risk analysts consider ourselves, with good reason, to be those experts who best understand probability and consequence (Kaplan and Garrick 1981; National Research Council 1983). We develop, refine, and apply methods to estimate risk (preferably with honesty about its inevitable uncertainty and interindividual variability—Morgan and Henrion 1990; Cullen and Frey 1999), and we opine on ways to reduce risk. With the aid of economists, we attempt to re-express risk-reducing benefits to society in monetary units that can be compared to estimates (Finkel 2014) of the costs of regulatory or “soft law” interventions.

But quantifying and manipulating numbers is not necessarily the same as understanding and appreciating those quantities. Public policies (be they new actions or policies that favor inaction and the maintenance of the status quo) affect society *because* they affect each individual within society. But while experts may rather easily understand and appreciate the import and implications of *collective* facts such as “1000 statistical lives could be saved for a collective expenditure of \$1 billion,” it is much less easy to truly appreciate individual estimates such as “an additional 1/100,000 chance of premature death,” or “reducing this toxicant will cost each citizen \$500/yr. in increased consumer prices.” To any affected person, more risk is *a priori* worse than less risk, and more cost worse than less cost, but beyond that, *comparing aggregate costs to monetized benefits tells us nothing about how each individual is affected.*

This article opens a door to information that could further improve cost-benefit analysis (CBA), by giving analysts and decision-makers information about the significant qualitative and quantitative differences between over-aggregated risks (and economic harms) *versus* individual welfare. And rather than inferring or speculating about how welfare is experienced by affected citizens, here we directly elicit citizens’ views about welfare effects that cannot adequately be captured by “total cost” or “total benefit” estimates.

To date, experts, pundits, and decision-makers have invoked words such as “trivial” and “insignificant” to refer to risks (and far less frequently, to regulatory costs) that *they* deem too small to matter, and words such as “unacceptable” to refer to effects *they* deem too large not to ameliorate. The quantitative risk analysis (QRA) literature, and the pronouncements of risk-reduction agencies and individual “experts,” are rife with statements that some particular risk is so small that it can be ignored or that it is actually indistinguishable from zero risk; in either

case, that probability is regarded as “*de minimis*” (Comar 1979; Fiksel 1985; Adler 2007).¹ Both the literature and policy statements are also replete with references to particular risks so large as to be “intolerable” (Sewell and Whyatt 1989; Boudier, Slavin, and Löfstedt 2007). And we assert as a truism that the amount of money a citizen must pay (or not receive, which is equivalent) so that a policy can take effect must *also* fall into one of these three parallel regions: an amount of cost so small it can be ignored, an amount of cost so large that it should not be tolerated or contemplated as a result of a policy, or any value in between these two “goalposts.” As we will discuss below, economists are far less prone than risk scientists to even mention the existence of “*de minimis costs*” or of “intolerable costs,” but this does not diminish their importance. This particular lapse is odd, likely ironic, and perhaps hypocritical, given that our field is heavily influenced by economists, for whom “consumer sovereignty” is supposed to be a paramount value (Acland 2022)!

But none of these terms have a generally-accepted meaning, and all previous attempts to define them quantitatively have failed to use the most basic tool—citizen elicitation—to estimate them with some respect for objectivity and for the basic premise that public value judgments should be informed by public values. Risk analysts and economists have instead preferred to **infer or impose** the locations of these boundaries via indirect, volatile, and sometimes irrelevant proxies. In fact, all the lip service given to *particular* very small and very large risks (and costs) may have lulled our field into the false conclusion that researchers, practitioners, and policy-makers *have the first idea what these dividing lines generally mean, where they fall, and how to make societal decisions in that vast majority of cases where risks and costs to some individuals are intolerably large while others are facing risks or costs that are trivially small.* Our risk management policies have perennially failed to act on the basis of these “bright” (or “fuzzy bright”) lines; we utter adjectives like “insignificant” and “unacceptable” but we fail to downweight the former or persevere against the latter.

This article reports on a systematic, citizen-centered approach to define and quantify the “four corners” of cost-benefit boundary-setting: where *de minimis* risk (and cost) and intolerable

¹ “De minimis” is Latin for “about trifles,” as in the hallmark aphorism that “the law is not concerned with trifles” (*de minimis non curat lex*). Therefore, we expect that the judicial system will refuse (in a principled way) to take up matters where the harm is so trivial as to be indistinguishable from no harm at all, or, at worst, from the kinds of harms humans silently bear incessantly.

risk (and cost) occur along the number line. In conducting these elicitations, we do not profess that the average (or any estimator of) citizen preferences is the end-all.² Just because there is a central estimate does not mean that this should be the focus when addressing distributional effects: the whole point of taking the distribution of benefits and costs into account is that some people's impacts from the intervention might count more than impacts on others. But these evidence-informed views are a superior source of evidence than arbitrary claims by experts. We cannot imagine how the economic model can respect "consumer sovereignty" without asking citizens to offer their judgments about risk and cost, and we cannot imagine optimal policies that don't in some way seek to respond to these preferences and views.

By shedding light on how society might better recognize *de minimis* harms and intolerable ones, we hope to enable decision-makers to do more than merely *remark* that some citizens may face trivially small effects while others may face intolerably large ones, but rather to *act* on those realizations. Currently, for all the thoughtful discussion of how different interindividual distributions of risk with the same total might be either "just" or "inequitable" (e.g., Robinson, Hammitt, and Zeckhauser 2016; Levy 2021; White House 2021), we still are virtually powerless to truly focus resources on intolerable harms and to avoid being distracted by trivial ones if we can't define when those two situations occur.

Simply put, if risks below *de minimis* levels are equivalent to zero risk, then we overreact to them when we treat them as mere "fractional lives"—as fodder for the standard way in which CBA deals with regulatory benefits [lives "saved" multiplied by the VSL (value of a statistical life)]. In the limit (assuming for the moment that a lifetime risk of 10^{-7} is below a *de minimis* level), it would be a mistake to regard a scenario in which all 330 million Americans face that small level as resulting in "33 lives lost," and a mistake to spend \$330 million to reduce those risks to zero (for a standard VSL of \$10 million per life). And yet regulators are loath to treat *de minimis* risks this way, preferring to *say* they are insignificant but actually treating them as the opposite.

² Eliciting citizen views directly is not the only way to learn about them. "Revealed preference" studies can sometimes infer these views by observing behavior, as when economists assert that the value of life can be inferred by estimating the extra wages paid to workers who "demand" them to compensate for specific excess risks of death. We encourage others to see if there exist any analogous ways to infer what risks citizens deem *de minimis* or intolerable by observing which risks they apparently pay zero attention to or seek to avoid at all costs. But this "stated-preference" experiment is how we've chosen to begin the effort.

At the other end of the spectrum of harm, we assert that (assuming for the moment that a lifetime risk of grave harm of 1/50 is intolerably high) it would be a mistake to regard 100 citizens facing that risk as “only” worth \$20 million (“2 VSLs”); if that situation is truly intolerable, it should not be traded off against monetary expenditure as if it merely took “1/50th of a life” from each grossly affected person.

The same logic, of course, applies to cost. If (again, for argument’s sake at this point) no citizen would notice or care whatsoever if a regulation caused the price of gasoline to rise by 1 cent per gallon (from, say, \$3.15 to \$3.16), the *de minimis* concept means that it would be an analytic mistake to ascribe a “cost” of \$1.23 billion each year to this effect (at 123 billion gallons purchased nationwide) (U.S. EIA, 2021), especially if that cost estimate caused a life-saving regulation to fail a cost-benefit test. Conversely, it is easy to envision a regulation that would drive a handful of businesses into bankruptcy and cause 100 workers to lose their careers with little prospects for retraining; estimating the cost of that policy as the lost wages alone would miss any notion that such life-changing costs may deserve special attention (Coglianese, Finkel, and Carrigan, 2014).

But with one and perhaps two obvious exceptions, U.S. federal and state regulatory-analysis policy clearly fails to treat trivial effects as truly trivial, and fails to treat intolerably large effects as truly deserving of special attention (Cranor and Finkel, 2018). The exceptions—the perennial special interest in small businesses (to the extent that many regulations simply exempt these firms regardless of the harms to workers or nearby residents)³, and the more recent special interest in very large cumulative risks to certain marginalized communities, in the name of “environmental justice”—demonstrate some willingness to deviate from the formulaic ways of estimating benefits and costs. But no one could seriously argue that these exceptions are applied uniformly or consistently, or are more salient than other situations where *de minimis* effects could be down-weighted or intolerable effects taken more seriously, but where neither refinement occurs.

This paper breaks new ground in two ways, each so fundamental to improving risk assessment and risk management that we are amazed (and not in a good way) to have this

³ In effect, “intolerable cost” is often currently defined as “whenever a special interest successfully argues that he/she/they shouldn’t have to pay that much,” which is of course (worse than) no definition at all.

opportunity so many decades into the ascendancy of QRA and risk-cost policymaking. First, we approach the question of how to estimate *de minimis* and intolerable harms in the most direct, and least-used, way of all: to carefully ask people to provide their own informed estimates thereof. Second, for as little attention as there has been to codifying the magnitude of *de minimis* and intolerable risks, the parallel notions of trivial and unbearable *costs* have virtually never even been mentioned, let alone quantified. But the entire foundation of CBA as the dominant policy-analysis method rests on the premise that risks and costs are *comparable and can be traded off one for the other in policy*. We can't possibly believe we are making benefits and cost commensurable if we don't at least think about whether some costs are so large as to be intolerable, or so small as to be infinitesimal no matter how many people happen to face them.

2. Clarifying Terminology:

2.1. *Definitions*:

Before proceeding to discuss the scanty previous literature, and our current methods, results, and their implications, we need to carefully explain some recurrent foci of this paper, and terms used herein.

- We construe “risk” and “cost” in what we consider to be their original meanings: phenomena that affect individuals, and *therefore* affect populations, rather than vice versa. Risk as an individual probability of harm, or as the extent of harm to a population, are two sides of the same coin, and were it not for the general case where individual risks are unequally distributed, there would be no need to emphasize one definition over the other.⁴ Public health does need to keep one eye fixed on the aggregate consequences of exposures to dangers and to the costs of reducing them. However, the concepts of *de minimis* effects and intolerability begin with *exquisitely personal appraisals of the kind we elicit here*, in part because “statistics are merely aggregate numbers with the tears wiped away.”⁵ We are not downplaying the importance of population risk and cost by emphasizing personalized effects here; rather, we are implying that some risks and costs

⁴ In the specific but rare case of perfectly equal distribution, the “body count” and the individual risk are just scalar multiples of each other, akin to measuring weight in milligrams versus kilograms.

⁵ This quote is attributed to pioneering asbestos researcher Dr. Irving Selikoff.

to society are paramount (or trivial) *because* of their aggregate effects upon each individual.

There is a prominent literature about how laypeople perceive different magnitudes of population consequence (Västfjäll and Slovic, 2013), so a different study than this one could certainly try to pin down further what numbers of “statistical lives lost nationwide” or of total regulatory cost laypeople deem *de minimis* or intolerable. There are also a few papers (e.g., Milvy 1986, Finkel 1992) proposing ways to judge policies simultaneously by how they reduce population consequence *and* how they reduce the maximum or the average individual risk (or cost), balancing the two goals according to ethical judgments about the importance of each. But the whole point of exploring effects first on individuals is to allow people to understand and make decisions based *either* on judgments about the benefits of policies to individuals or to populations—the latter naturally flows from summing up the former, but in the reverse direction there may not be enough information to derive individual prospects from population consequences. By exploring the four corners of individual risk and cost, rather than population effects, we are also comporting with the view of the U.S. Supreme Court the only time it ventured into the QRA domain. In the 1980 “Benzene” case (Industrial Union Dept., AFL-CIO v. American Petroleum Institute, 1980) the Court instructed OSHA that it must demonstrate that a risk is “significant” before it can regulate it, and it must not regulate so stringently that the risk falls below *de minimis* levels. The Court’s only quantitative guidance *came in units of individual risk*: that a lifetime excess risk of death⁶ above 10^{-3} must be “significant” and that a risk below 10^{-9} could not possibly be significant. If the Court had instead told OSHA that it must regulate hazards whose death toll exceeds X persons/year, and could not regulate so as to reduce the toll to below Y deaths/year, the landscape of risk regulation might be different today, but we assume this choice of units was deliberate on the Justices’ part (see Adler 2005 for a very similar interpretation of this case). In the very few statutes where Congress itself has instructed an agency (usually EPA) what risks must be reduced and what risks should not be reduced further, it has also opined in units of individual, not population risk. The Clean Air Act Amendments of 1990 specify that EPA should consider regulating a source category if the maximally exposed individual would face a risk of greater than 10^{-6} —*not* that it should do so if more than X statistical lives would be lost absent regulation—and similar provisions in the Food Quality Protection Act and other laws construe risk similarly.

- We explicitly ask subjects to consider only risks and costs that are *involuntary*, as opposed to those resulting from personal choices (e.g., the fatality risk of skiing, or the

⁶ In this article, we are concerned only about *de minimis* risks when the consequences are maximally severe (premature death). There is a separate literature about trivial *consequences* (e.g., what amount of cognitive decrement from lead exposure is too small to concern even those affected?).

cost of a vacation). In that way, we can strive to obtain responses that truly represent public perceptions of what makes a risk that is imposed upon some or all either *de minimis* or intolerable—too small to think about *even though* the risk is not freely chosen, or too large to tolerate *even if* the subject may voluntarily choose to bear other risks of similar or greater size. Similarly, we seek to elicit responses regarding what amount of regulatory cost is trivial or intolerable even though (in the former case) the subject may perceive no benefit from the regulation, or (in the latter) the subject may choose to make purchases greater than the amount s/he would not want to face as part of a social program to reduce some hazard. Again, we are trying here to shed light on when a generic risk or cost becomes “too small” or “too large,” *without* specifying any other attributes of the effect other than its involuntary nature. Subsequent studies could then subdivide each of these four “goalposts” by other salient qualities of the effect (type of health hazard, payment via tax or price increase or some other means, etc.).

- We also made clear to respondents that on the risk side of the ledger, we were only interested in varying the probability of harm, not the severity. So we deliberately made the bad outcome particularly vivid and severe (see Section 4.1 below). Subjects therefore provided data about the “goalposts” demarcating magnitudes of the probability of this fixed outcome, as opposed to some conflated metric such as “expected life-minutes lost” derived by multiplying probability and outcome. Future studies of this type could certainly provide complementary information on the probabilities laypeople regard as *de minimis* or intolerable for other outcomes (e.g., non-fatal injuries of various types).
- We are *not* looking here for the magnitude of a risk or cost that makes it “significant” or “acceptable.” The notion of “significance” has spawned a small literature (see below) that has not grown much in recent decades, but “significance” is a dividing line that by definition falls somewhere *above* our respective lower “goalpost” and *below* our upper one. And while “acceptable” is sometimes construed to be the obverse of “significant” (in other words, once a risk has been lowered so as to not be “significant,” that is synonymous with “acceptable”), we are not probing that either. These two adjectives are clearly context-specific: most obviously, a large risk can be “acceptable” if it provides ample benefits. In this article, we are probing *context-independent* bright (or “fuzzy bright”) lines. Here and for the remainder of the article, we will use the subscript “D” as an estimate of the magnitude of a risk or cost at and below which it becomes *de minimis*, and “I” as an estimate at and above which it becomes intolerable. So this study divides all risks (or costs) into three regions: below D, between D and I, and above I. *By definition*, risks or costs below D are both “acceptable” and “insignificant,” and risks or costs above I are both “unacceptable” and “significant.” However, in the region between D and I the effects, though “non-trivial,” *may or may not be either “significant” or “unacceptable,”* and this paper does not offer judgments about these latter two

adjectives. We believe it can be of great value to understand better when *in all contexts* risks or costs are too small to be of any concern and when they are too large not to be of paramount concern.

- We will construe in this article the judgments of “too large” and “small enough” as two *separable and separate appraisals for risk and for cost*. Our online experiment very clearly separated the two elicitations, and made clear to respondents that they were not being asked to make any tradeoffs of risk for cost, but to ponder whatever first principles and/or ethical judgments make *either* risk or cost sufficiently low or intolerably high when viewed in isolation. As a result, we believe we have shed light on when a risk is so small that it can be ignored *even if it may also be trivially easy to eliminate*, and when it is so large that it is intolerable even if it may also be extremely costly to reduce. Similar descriptions apply on the other side of the ledger (a cost that is so small that it can be ignored even if it may be imposed in order to reduce a tiny risk, or one that is so large that it cannot be tolerated even if it may be needed to reduce a very large risk).

2.2. Scope:

This paper, for reasons of length and complexity, leaves one set of contemporaneous experiments to a manuscript in preparation. Stopping here, as we do, with estimates of each of the “four corners” (including their interindividual variability) will allow society (see Section 8.6 below) to adapt risk management policy in innovative ways. However, dividing risks and costs into three qualitative regions (below *de minimis*, above intolerable, and everything in between) doesn’t shed light on the nature of these ubiquitous “in between” effects, on either side of the cost-benefit ledger. In each of our experiments here, we also asked respondents to offer a relative “quasi-willingness-to-pay” amount to avoid each of four levels of risk or cost between their own user-defined values for the lower and upper endpoints; we will report on those results, and what they imply about the non-linearity of harm as individual cost or risk rises from acceptable *towards intolerable levels*, in our subsequent publication.

3. Previous Approaches to Estimating *De Minimis* or Intolerable Harms

Of the “four corners” estimated in this set of psychometric experiments, we are well aware that prior investigators have speculated about the magnitude of one of the four in particular—that of *de minimis* risk (hereinafter, R_D). In this section, we will briefly catalogue the various prior methods used to make statements about R_D , discuss the much more sparse efforts to reify in any way the other three corners, and then summarize reasonable past estimates of each corner before we turn to our new results for all four.

3.1 Past Discussions of De Minimis Risk:⁷

Over the past 40 years, various investigators have made pronouncements about the approximate value of R_D using a surprising variety of disparate approaches to the task. This brief catalog of approaches shows how many ways researchers can opine about R_D , *without ever asking citizens for their views about this quintessentially subjective and value-laden concept*. In the past, scholars have opined about R_D by invoking⁸:

- *Appeals to folklore.* Although the original source of this quip is unknown to us, we’ve often heard risk analysts say that 1/million was originally attractive as the benchmark for *de minimis* risk because “no one ever gives out the compliment that ‘you’re one in a hundred thousand!’” Indeed (Kelly, 1991), it does seem that no one in FDA or EPA at the time that 1/million gained currency can remember precisely how that number arose or was explained, although we hasten to add that sarcasm about the origins of 1/million in no way refutes the premise that it might *happen* to be the most reasonable estimate of R_D . Regardless of the formal or informal reasoning first applied, it is clear that beginning in 1975, FDA was using the modified Mantel-Bryan (1961) suggestion for 1/million as *de minimis* (Rodricks 1988).⁹
- *Appeals to personal opinion.* It would be impossible to catalog the vast array of statements in the literature favoring a *de minimis* probability other than 1/million, but for one recent and representative example, Prof. Joseph Allen recently told a reporter for *The*

⁷ We note particularly three excellent books published between 1976 and 1987—Lowrance 1976, Fischhoff et al. 1981, and Whipple 1987—devoted to discussions of *de minimis* risk. However, none of the chapters in any of these books attempt to estimate R_D using elicitations from laypeople, as we do here.

⁸ Besides these single studies we cite, there are several excellent reviews of multiple studies here, notably Hrudý and Krewski 1995, Hunter and Fewtrell 2001, Rae 2007, and Roy and Kshirsagar 2021.

⁹ FDA has described very small risks as *de minimis*, which of course is consistent with using a higher threshold (1/million) as an upper limit for this concept. Blumenthal (1990) notes that the estimated cancer risk for D&C Orange #17 was one in 19 billion, which FDA described as *de minimis*.

Atlantic (Khazan 2022) that “If you are vaccinated, boosted, and wearing a well-fitted N95 or similar indoors, your risk [from Covid-19] is extremely low... there’s not much else in life that would have as low a risk as that. I would qualify your risk as *de minimis*.”¹⁰ Near the other end of the spectrum (as mentioned above), the Supreme Court opined in 1980 that while OSHA was free to choose a higher dividing line for *de minimis* risk, it would not be able to regard lifetime risks of 1×10^{-9} or lower as worthy of any attention. The Court did not defend its choice of 1/billion as R_D here with any data or analogies.¹¹

- *Appeals to ethics regarding probability and consequence.* As part of a landmark paper that was among the first to try to ground *de minimis* risk both in terms of individual probability and the population death toll, Milvy (1986) began from the premise that 3 or 4 excess deaths per year in the U.S. (the result of a 10^{-6} lifetime risk applied to the entire nation) was “very small” compared to the national death toll and the number of deaths from any of the major causes thereof. Although most of his article derived a formula for R_D ($R_D = 0.015 * P^{1/2}$, where P is the size of the exposed population) that could be used to relax the *de minimis* criterion for exposures that affect fewer people, he used both 1/million and higher values to argue that his formula is ethically reasonable.
- *Post hoc appeals to how agencies actually behave.* A landmark paper in the small literature on R_D was written by Travis et al. in 1987. They arrayed 132 decisions by federal regulatory agencies by individual risk, population consequence (death toll), and by whether the agency took action or declined to do so. Figure 4 in that article demarcates the region presumed to reveal where *de minimis* falls, because no agency ever chose to regulate once these risks were estimated as such. For large populations, the *de minimis* individual risk was 10^{-6} , but rising to 10^{-4} when the exposed populations were on the order of thousands rather than millions.
- *Appeals to “trivial additions to background.”* Peterson (2002) discusses this notion extensively, referencing several authors who have argued that because we tolerate

¹⁰ Using January 2022 data from Oxford’s Global Change Data Lab (<https://ourworldindata.org/grapher/united-states-rates-of-covid-19-deaths-by-vaccination-status?country=~50%2B>), the risk of death for a boosted individual was about 0.3 per 100,000 per week. Using a 95% reduction to approximate the effectiveness of a well-fitted N95 respirator, this equals roughly a 7.5×10^{-6} *per year* fatality risk. It’s hard to extrapolate this mid-pandemic annual risk to a lifetime risk, but it is clearly *far* higher than 1/million/lifetime, so Allen is favoring a quite permissive definition of “not much else in life as low as that.”

¹¹ To be more precise, the language from the “Benzene” case referring to 10^{-9} used the example of “taking a drink of chlorinated water.” So it is possible that upon converting this one-time tiny probability to a lifetime one (say, by multiplying by 8 drinks/day and 25,500 days per 70 years, this *de minimis* pronouncement actually represented a lifetime risk of 2×10^{-4} —although this would be suspiciously close to the other risk value (10^{-3}) that Court used in the same sentence as unacceptably high. It’s probably more likely that the Court meant a 10^{-9} lifetime risk, in apposition to a 10^{-3} lifetime risk, and just explained the former example poorly.

(perhaps to the point of ignoring) particular levels of risk that are “the natural background,” that small *relative* increments of involuntary risk above the background could be considered *de minimis*. Both Rae (2007) and Roy and Kshirsagar (2021) discuss the German concept of “minimum endogenous mortality,” and recommend that risks below 5% of the (low) background mortality rate of children aged 5 to 15 should be considered *de minimis*.¹² Peterson references (rather negatively) pioneer risk analyst Alvin Weinberg’s argument that “in spite of the ubiquity [of the background risks], humans have survived” as his justification for ignoring increments above a fraction of the background. Byrd and Lave (1987) note that the U.S. Nuclear Regulatory Commission had several years earlier defined a “trivial level” of risk as “0.1% of the sum of prompt fatality risks resulting from other accidents to which members of the U.S. population are generally exposed.” Regardless of the multiplier or the type of background rate chosen, these appeals rely on the notion that R_D derives its meaning not from the absolute chance of harm, but from whether additional harm meaningfully changes the prospects that a citizen already faces.

- *Appeals to the fluctuations within the background.* An approach that takes the background into account, but that doesn’t insist that “a small fraction of a large risk must be trivial,” relies instead on the idea that the layperson might well ignore an increment of risk so small that it *could just as well be* “noise” within the background. For example, Spanger (1987) suggested that while the natural background of ionizing radiation is roughly 300 millirem, this amount varies spatially by about 100 millirem. Arguably, therefore, while the background risk itself may or may not be trivial, moving to an area with 100 millirem more or less than one faced before might well be *de minimis* because one would be trading one background level for another.¹³ Foreshadowing the analogy to *de minimis* cost may be helpful here: while the full price of a gallon of gasoline may well be a tangible expense for some people, it would be unusual for someone to notice a 1 cent/gallon increase from one week to the next, or to wait to fill the tank in the hope that the price would return to the prior level.
- *Appeals to what is “observable.”* A common theme in the small literature on R_D is that the reason a small increase in a larger probability is *de minimis* is because said increase would “be lost in the noise of background variation.” Byrd and Lave (1987) used these words to conclude that an annual risk of 10^{-5} (a lifetime risk of 7×10^{-4}) is *de minimis*. Others have invoked the rule-of-thumb that only a doubling (or more) of the background

¹² This background rate is roughly 2×10^{-4} /yr in the U.S. and E.U., so 5% of that annual risk, converted to a 70-year lifetime, would imply a rather high estimate of R_D , at 7×10^{-4} .

¹³ As Richard Wilson wrote in 1988 (supporting a value of 7×10^{-5} as *de minimis* because it is about $1/10^{\text{th}}$ of the *variability* in the natural background exposure to ionizing radiation), “I know of no one in the world who alters, or even chooses, his place of residence or his lifestyle to keep his natural radiation dose level low.”

risk in an epidemiologic study (Carruth and Goldstein 2001) describes a functionally meaningful (as opposed to statistically-significant) increase, to imply that estimated relative risks of less than two-fold are *de minimis*.¹⁴ Still others (see, e.g., Peterson 2002) have suggested that the amount of excess risk associated with an undetectable increase in a particular *exposure* (Peterson used the example of a 10 µg/dL increase in blood lead level, which may have been near the detection limit at the time) should be deemed *de minimis*. The logical flaws in these approaches are, we hope, obvious: among other things, they hold protection from involuntary risks hostage to a notion of whether the beneficiaries could “know” they were at risk (the “tree still falls” whether or not someone is there to hear it fall). More worrisome, this kind of approach implicitly deems larger absolute risks to be acceptable when the disease endpoint is common rather than rare; a 90% increase from 1 in 1000 to 1.9 in 1000 in a common but grave disease could be deemed *de minimis* (despite its absolute risk increment of 9×10^{-4}), while a much smaller absolute excess risk (say, a 10-fold increase from 10^{-8} to 10^{-7}) in a rare disease could be deemed important.

- *Appeals to what is “practicable.”* Many authors, including Bowles (2007), have discussed “ALARA” (“as low as reasonably achievable”) or “ALARP” (“... practicable”) as a way to benchmark *de minimis* risk: if means do not exist to reduce a risk further, it can perhaps be regarded as *de minimis*. As with the above approach, this rejects the notion that “too small to consider” has a fixed meaning, and so it courts over-regulation (when an exposure happens to be easily controlled, even if its magnitude is trivial) as well as under-protection (when a large risk cannot “reasonably” be diminished). The latter problem could be quite significant in cases where a product or process could be banned, but society regards that response as “unreasonable” or “impracticable.”

For completeness, we emphasize that a few scholars have utterly rejected the idea that *any* small risk can be ignored as *de minimis*. Lundgren and Stefansson (2020) opined that R_D is “deeply flawed from a theoretical perspective,” while Adler (2007) wrote that R_D has “no basis in ideal moral theory.” Lundgren and Stefansson, in line with Peterson (2002), agree that the “small relative to background” and “below detection limits” approaches to estimating R_D are illogical and morally suspect, but they go further to claim that because any specific number for R_D is “arbitrary,” that no such number can be valid. But the entire *raison d’être* for our field of risk assessment and management is that sensible, evidence-based, and publicly-responsive judgments about risk are the *least* arbitrary ways to discuss and make the “tragic choices” (Calabresi and

¹⁴ A related approach was discussed by Hunter and Fewtrell (2001), who suggested that society could look at all cases of a particular disease (they used gastroenteritis as an example), deeming as *de minimis* any exposure that contributed to less than some percentage (say, 5%) of all cases.

Bobbitt, 1978) that will be made consciously or tacitly no matter what. The fact that a speed limit of 5.5 mph or 550 mph might be as “arbitrary” as 55 mph is not an argument for no limit at all.¹⁵ Adler claims that R_D ruins a consequentialist view of welfare, but relies on a definition of welfare here that only considers the death toll as relevant, not the probability of death to any individual. While we agree that rounding down to zero those individual risks below R_D will bias the estimate of the death toll, we repeat that we regard infinitesimal changes in a person’s life prospects as irrelevant to welfare. Again, the analogy to cost may help: we agree for argument’s sake that the nation is worse off having spent \$10,000 in tax revenue to build a “bridge to nowhere,” but we simply can’t accept that the expenditure of \$0.00003 per capita makes any individual American worse off. Adler also argues that allowing for *de minimis* risk ruins a rights-based ethic (noting that someone who forces another to play Russian Roulette with 10^{-7} odds of death could in fact end up murdering that person), but we again reply that the whole point of R_D is to distinguish between meaningful and meaningless odds.¹⁶

The startling aspect of these many past attempts to estimate—or to eschew—*de minimis* risk is that none of these authors has considered what citizens themselves have to say, despite the centrality to economists of individual-specific utility and consumer sovereignty. In the environmental risk literature, we’ve found only one peripheral (and “rough and ready”) reference to actual citizen preferences: Byrd and Lave (1987) offer that a lifetime risk of 10^{-3} is insignificant because “many people tend to treat all risks smaller than 1 in 1,000 as equivalent.”¹⁷

¹⁵ Lundgren and Stefansson make two other unusual arguments that we disagree with: 1) that because “one person’s sure death cannot be *de minimis*,” the entire concept is flawed. We respond that R_D is a probability for all, not a certainty for any one; and (2) that if the world population was 10^{20} , a risk of 10^{-12} might not seem so negligible. We agree, but point out that the *actual* world population is 8×10^9 , with obvious implications for R_D .

¹⁶ The claim that the probability of a given harm could even have its own “disutility function” is controversial. Some observers (Pratt and Zeckhauser, 1987) claim that this is nonsensical—that only the *outcome* (in this article, death) can have disutility, but the probability of the outcome cannot. By this view, a 1-in- X chance of harm has exactly $1/X^{\text{th}}$ the disutility of the harm itself, for all X ; different people can have different views about (different willingnesses to pay for) the harm, but not about the probability (see EPA 2011, p. 6: “it seems unreasonable... that an individual would value a second risk reduction of Δp much less than an initial reduction of Δp ”). This view, of course, would not allow even for the very notion of *de minimis* risk (a very small probability of a given harm is just a very small fraction of whatever the disutility of the bad outcome is), or of intolerable risk. It also reflects an expected-utility view of human perception, a view that has been waning for nearly a century (Allais, 1953) and that continues to be supplanted by richer depictions such as regret theory (Bell, 1982) and prospect theory (Tversky and Kahneman, 1992; Booij and van de Kuilen, 2009).

¹⁷ One similar invocation of citizen preferences is much older than Byrd and Lave’s article. According to Hey et al. (2010), the Comte de Buffon argued in 1777, as an attempt to use R_D to “resolve” the “St Petersburg Paradox,” that a rational person would offer no more than \$13 to play this infinite-expected-value game. Buffon argued that since

The experiments reported in this article represent (or at least strive to be) the opposite of such top-down pronouncements about what people actually think and believe.

We will present in Section 5.4, in ascending numerical order, some estimates of *de minimis* risk from the articles cited above, after presenting the results of our experiment, so we can put the findings in context of prior estimates.

3.2 Past Discussions of Intolerable Risk:

In stark contrast to the plethora of articles that at least touch on *de minimis* risk issues, there are very few discussions of intolerable risk in the literature. To be sure, the explorations of “acceptable risk” often explicitly or implicitly opine that any probability higher than this particular dividing line is “unacceptable,” but we draw a sharp distinction in this article between this adjective and “intolerable.” The modifier “unacceptable” clearly connotes something context-specific: an amount of risk that should be reduced either because of the benefits of doing so exceed the costs of control, or because the source or type of hazard is particularly objectionable. “Intolerable,” however, means a *context-independent* level of risk that no citizen should have to bear (again, there may be some intolerable risks that cannot be reduced, but this doesn’t change the meaning of the term). One excellent definition of R_I for our purpose comes from the U.K. Health and Safety Executive (2001): “all individuals have unconditional rights to certain levels of protection... In practice, this often converts into fixing a limit to represent the maximum level of risk above which no individual can be exposed.” Therefore, while there are various approaches (see above) to thinking about R_D , intolerable risk is fundamentally a subjective and ethical value judgment—but one that can be probed via the kind of elicitation we undertook here.

However, the few references to the actual size of R_I in this literature tend to construe what is “intolerable” in terms of the frequency of fatal accidents that kill N people, not in terms of individual probability of harm. For example, Roy and Kshirsagar (2020) display a variety of “F-N curves” (fatalities and numbers), which regards as “intolerable,” among other

a 56-year-old man “ignores the possibility of dying in the next 24 hours,” which had a probability of 1/10,189 according to mortality tables of the time, events with less than 1/10,000 probability could be ignored.

combinations, an accident that kills 100 or more people and whose frequency is more than 1/1000 per year. This would actually translate into a strikingly small individual risk (below 2×10^{-8} per lifetime) for any random American, so there must be something about the aversion to large-N events that doesn't translate well to judgments about what personal probability of grave harm is ethically intolerable.

One isolated use quantifying “intolerable risk” is found in the title of a 1989 report from an environmental group (Sewell and Whyatt), which played a major role in causing the voluntary withdrawal of the growth hormone daminozide (“Alar”) for use on the nation’s apple crops. The Natural Resources Defense Council estimated the lifetime cancer risk for a high-end consumer of apple products containing daminozide to be 2.5×10^{-4} (see Finkel 1995 for a detailed discussion of the basis for this risk estimate), and deemed that probability “intolerable.”

The only analog to the canonical value of 10^{-6} for R_D we’ve found for intolerable risk can be found in Aven (2007), who describes an annual risk of 0.1% (10^{-3} /yr, or 7×10^{-2} per lifetime; one chance in 14) as “the traditional level.” Roy and Kshirsagar endorse this choice, going further as to suggest that the 10^{-3} /yr level is generally applied to workers, while a stricter criterion $1/10^{\text{th}}$ as large (that is, 7×10^{-3} per lifetime, or one chance in 143) applies to the general public.¹⁸

Interestingly, Aven states that “adopting the traditional level 0.1%, it is known from many years of experience of using this criterion that it is met for most or nearly all types of activities in the Western world.” This is overly sanguine. There are at least 10 common occupations in the U.S., including logging, roofing, and underground mining, where annual on-the-job fatality rates (AFL-CIO 2022, p. 18) exceed 2×10^{-4} . Risks of occupational disease often exceed even this large value; the Center for Public Integrity (Qiu and Hopkins 2015) estimated

¹⁸ The upper end of the Supreme Court’s pronouncement in the “Benzene” case is 10^{-3} per *lifetime*; although there is very little case law suggesting that OSHA has an absolute duty to regulate risks above 1/1000, by leaving it to OSHA to determine where *below* 1/1000 the boundary between “significant” and “insignificant” risk must fall, the Court may have implied that its upper benchmark for risk is *prima facie* “intolerable.”

lifetime cancer risks associated with OSHA's Permissible Exposure Limits (PELs) and found many cases where these risks exceed 10%.¹⁹

3.3 *Past Discussions of Trivial and Intolerable Costs:*

We frankly find it remarkable, and revealing, that contrasted with all the attention paid in the risk analysis literature and regulatory policy to defining *de minimis* risk (and at times disrespecting citizens' claims for relief from such risks), there has almost literally been zero interest in the parallel concept of *de minimis* regulatory cost. Internet searches for "*de minimis* cost" almost entirely yield references to a 1977 Supreme Court decision (EEOC, 2021) that defined this concept as an aid to judging whether an employer can refuse to accommodate the religious beliefs of its employees (the Court offered "more than *de minimis* cost" as an interpretation of the statutory term "without undue hardship"). There have been a very few papers discussing whether very widely diffuse health/environmental regulatory costs could be ignored. Driesen (2004) offers that "often costs imposed on companies become distributed so widely that they have little real impact on human lives," using as an example a rule that cost the nation \$100 million per year, consisting of 100 million increments of \$1 costs upon individuals. In the "implications" section of this article, we quote Sunstein (2021) making a similar argument using a \$4 annual per capita cost. *But we are unaware of a single recommendation, let alone a single Regulatory Impact Analysis*, that has ever taken a total cost estimate and reduced it by rounding to zero (or downweighting) the portion of total cost that imposes *de minimis* economic impact on portions of the population. Needless to say, there have been no serious attempts to quantify where C_D might fall.

The public discussion of intolerable costs is more prominent, as seen especially in the extensive attention regulators (must) pay to economic effects on small businesses.²⁰ *However,*

¹⁹ For example, the OSHA PEL for trichloroethylene (TCE) is 100 ppm. EPA's inhalation unit risk factor for TCE is 2.3×10^{-5} per (part per billion).

²⁰ Since 1996, U.S. federal regulatory agencies must separately analyze the economic effects of their rules on small businesses, and three major agencies (EPA, OSHA, and the Consumer Financial Protection Bureau) must invite a panel of 15 small business representatives to jointly develop a report recommending special accommodations for this constituency. This requirement gives "intolerable cost" to one subgroup a favored (early) place in the regulatory development process, and there is no parallel consideration extended to those who bear intolerable risk

we are unaware of any attempts to codify or “guesstimate” how large a cost to an individual or a company must be in order to qualify as “intolerable.” The only reference point on which economics seems to sometimes rely is the event of an individual losing her job as a result of a regulatory decision. While unemployment clearly qualifies as a significant, perhaps an intolerable life event (see chapters in Coglianesi, Finkel, and Carrigan 2014 summarizing the literature on the welfare and morbidity effects of unemployment), it is difficult to ascribe significant numbers of net job losses (ibid.) to any particular regulation or even to environmental/health/safety regulations as a whole. Nevertheless, we will argue below that an improved version of CBA could tally net job losses as one proxy for “extent of intolerable cost.”

3.4 *Summary of Past Discussions:*

If it seems worthwhile to consider removing the straitjacket that forces decision-makers to treat all probabilities of harm as scalar multiples of each other, and all costs to individuals as if “dollars and utility are equivalent,” then the first step must be to probe what magnitudes of risk and cost demarcate the three regions of *de minimis*, intolerable, and points between. But for all of the invocations of terms like “trivial” and “unacceptable,” there exists no evidence-based discussion of where *de minimis* risk falls, and virtually no discussions at all about the other three corners.

Perhaps more astounding, few if any of the pronouncements about any of these four benchmarks has ever followed from any attempt to elicit citizens’ views. Instead, the boundaries have been invoked based on folklore, analogy, expediency, or *post hoc* rationalization. We offer alternatives.

4. Materials and Methods:

4.1 *Scenarios Used in Our Experiments.*

(perhaps caused by the harm-promoting exemptions given to small business!); see Finkel 2011 for a proposal in this regard.

In designing the elicitations reported on in this article, we took particular care to construct the survey so as to give subjects the best possible opportunity to understand and appreciate the probability of harm and its severity, without taxing their patience or abilities. Since the goal here was to elicit estimates of where risk becomes “*de minimis*” or “intolerable,” we needed to make sure subjects understood small and large probabilities, and understood costs in context of familiar expenditures. We also wanted to ensure that subjects would react as realistically as possible to a risk (or cost) scenario that clearly was a thought experiment rather than a real dilemma or anything that could be too specific to be broadly applicable—but still visceral and personal enough to elicit thoughtful reactions.

We first asked subjects to pledge to answer the questions truthfully (Jacquemet et al., 2013). We then provided one screen of introductory material about the experiment—basically, introducing the purpose of eliciting the subject’s opinion about what constitute “small” and “large” probabilities of grave harm, or of small and large amounts of regulatory cost. We made clear to the respondents that the purpose of what followed was “to help decision-makers respond better to how citizens think and feel about the varying probabilities of harm citizens face, rather than more simply comparing the total consequences of a hazard (e.g., the number of lives lost) to the total costs of controlling it.”

The risk scenario then followed. In order to isolate as much as possible the harms being considered from any ancillary benefits, we posed the basic risk scenario as one where the hazard was an unnamed and hypothetical consumer product (one “useful to you but not indispensable”) that was suddenly discovered by science to emit trace amounts of a harmful substance known to cause a rare, rapidly-fatal, and incurable disease. The product was left “unnamed” to avoid any particularized associations subjects might have, but we did suggest that subjects might think of the product *as if* it was their lawnmower—if the engine was found to produce trace amounts of some deadly emission, safer alternatives would be available (push mowers) but cumbersome.²¹ If we had allowed subjects to imagine a product that had little or no usefulness, risk tolerance would presumably drop at both the R_D and R_I ends of the spectrum (why accept *any* risk for zero benefit?); on the other hand, if we allowed them to imagine an indispensable product, tolerance

²¹ A reviewer helpfully pointed out that some subjects may not see a lawnmower as at all useful in their life, while others might regard a power mower as necessary and a push mower as an unacceptable though safer alternative. See Section 7 (“Limitations”).

would likely rise. By describing the product this way, we hoped subjects would give responses closely aligned with a “pure” version of risk tolerance, without making the responses in any way beholden to benefits of a particular size or type.²²

Then, in order to make the outcome associated with the small probability as realistic and dire as possible, we asked subjects to imagine they were waiting in a doctor’s office for medical test results, and that if the risk had manifested as harm, the doctor would enter the room and tell the subject that there was no doubt that the substance had caused her/him to contract a disease that would cause their death within 3 months. We asked all subjects, regardless of their age, to imagine themselves healthy and at age 35, in order to avoid the possibility that very sick or very elderly subjects would not regard this outcome as maximally dire. We provided a photo of a grim-faced doctor and a script for them to read to themselves.²³ Because we emphasized that subjects should think about the random chance of getting the bad news as “your number has come up” as in a card or a dice game, the experiment construed risk as a once-in-a-lifetime (not a repeating annual) probability.

For the parallel study probing reactions to individual cost, we again needed to set up a scenario where subjects could choose amounts of *regulatory* cost that were trivial or intolerable without reacting to a specific cost. So, we merely specified that the cost was being incurred in order to completely control emissions from some industrial process that, if left uncontrolled, would result in roughly several thousand Americans dying prematurely each year from a disease it caused. We left the harm vague, and merely said it was “about the same number of deaths as those due to drowning,” in order to connote a cost that was neither amassed for some trivial purpose (thereby depressing the amounts of cost that subjects would likely regard as *de minimis* or intolerable) nor for something so dire that “money is no object” (thereby inflating responses

²² Another important implication of the way we designed this scenario is that all of the data herein should be interpreted as pertaining to *a single narrow matter like the costs and benefits of one particular product, hazard, or exposure*. We are not seeking estimates of “*de minimis* cumulative risk from all air pollutants considered together,” or “what aggregate risk from all EPA-regulated substances is intolerable?”, or “how much should a citizen never be expected to pay for all public policies combined?” Future research could certainly define the “four corners” in more expansive ways.

²³ We asked each subject to provide an alias last name for themselves, just to personalize the script (while keeping their real last name hidden). So if someone said their last name was “Johnson,” and indicated they were female, a photo of a doctor talking to a woman would appear on-screen, and the script would start with “I’m so very sorry, Mrs. Johnson, to have to tell you this awful news, but...”

for both benchmarks). We also told subjects to imagine that the pollutants would be controlled in ways that would add some amount to the prices of common goods s/he and all other citizens would pay, goods such as cell phones or automobiles.

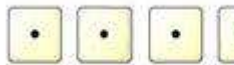
The actual risk and cost experiments used can be accessed in an interactive form at <https://doi.org/10.3886/E175661V3-140123> (see the two URLs at the very bottom of this “OpenICPSR” webpage); no data will be collected if readers wish to “fill out” the static survey(s) and familiarize themselves with the questions we posed.

4.2 Imparting Baseline Knowledge about Probability (and about Cost).

In both experiments (especially in the risk elicitation), we faced the daunting challenge of trying to ensure that subjects truly understood the nature of probability or cost, both in absolute terms and in context of more familiar examples, yet without overwhelming them with information. We struck this delicate balance by providing subjects with information in this order: (1) a table (reproduced here as Figure 1) showing how a wide range of common probabilities (e.g., 1 in 10; 1 in 500; 1 in 1000; 1 in one million) could be realized as specific combinations of drawing standard playing cards or rolling one or two dice, and also corresponded to the lifetime odds of specific causes of death, such as fatal lightning strikes or accidental death in a “bouncy castle”; (2) two “icon arrays,” showing what one chance in one million and one chance in 10 looked like in an array containing 1 million grey and one (or 100,000) red pixels; and (3) photographs of an antique revolver containing 10 chambers (instead of the more familiar six), and asked subjects to imagine playing “Russian Roulette” with one gun, or by choosing one gun at random from a group of ten (a fatality risk of 1/100 since only one gun was said to have one bullet in it), or by choosing one gun from a photograph containing 100 guns (1/1000 risk).

FIGURE 1: Part of the online experiment, introducing subjects to basic probabilities and actuarial risks

Common Probabilities with Dice, Cards, or Common Causes of Fatal Injury			
Probability	Dice	Cards	Common Causes
1 in 10 chance (this is the same probability as 100,000 in a million) 10%	Rolling <i>either</i> a total of 2 OR of 3 with 2 dice [actually 1 chance in 12]	Turning over one card and finding it is either an ace (of any suit) or the Jack of Hearts (5 cards out of 52, or almost exactly one chance in 10)	
1 in 100 chance (10,000 in a million) 1%	Rolling a total of 4 with THREE dice (1, 1, 2 in any order) [actually 1 in 72] 	Dealing 4 cards in a row of the same suit, ANY suit [actually 1 in 95] OR Dealing 3 spades in a row (1 in 77.3) 	
1 in 500 chance (20,000 in a million) 0.2%		Dealing a "flush" of 5 cards in a row of the same suit, any suit [1 in 504]	
1 in 1,000 chance (1,000 in a million) 0.1%	Rolling four ones in a row [1 in 1296]		Fatal drowning in pool or hot tub [1 in 2400, assuming the risk for kids under 15 persists for a 75-year lifetime]

1 in 10,000 chance (100 in a million) 0.01%	Rolling five ones in a row [1 in 7,776] 	Dealing 4 of a kind (4 aces or 4 twos or 4 threes, etc.) in a row  [actually 1 in 20,825]	
1 in 100,000 chance (10 in a million) 0.001%			Fatal dog bite [1 in 133,000], or lightning strike [1 in 80,000]
1 in 1,000,000 chance (1 in a million) 0.0001%	Rolling eight ones in a row [1 in 1,679,616]	Dealing either a "royal flush in spades"  OR a royal flush in hearts  [actually 1 in 1,300,000]	Dying in a "bouncy castle" [1 in 4 million]

We explicitly told subjects to imagine that their doctor was in the next room awaiting the test results, and if (for example) their probability of being given the rare disease was 1 chance in 77, then the doctor would actually have a deck of cards and would give you the dire news if he drew three spades in a row from the deck (an actual probability of $[(13/52)*(12/51)*(11/50)]$, or 1 in 77)). Finally, before beginning the interactive portion of the experiment, we asked people to ponder for a moment how they would regard each of these (broadly-dispersed) probabilities of the grave disease, with reference to examples in the “table of odds” we provided: 1 chance in 5 million, 1/133,000, 1/7,800, and 1/72.

For the parallel experiment on cost perception, our task was simpler, as dollars (unlike probabilities) are very familiar to Americans. However, we did provide a table containing nine examples of the amount of total annual spending (and per capita spending if the costs were borne universally) of some public projects (e.g., the \$55 billion annual budget of the U.S. Department

of Homeland Security, about \$180 per capita) or private purchases (e.g., the \$7 billion (\$25 per capita) Americans spend annually on potato chips). We also asked two factual questions as a rough proxy for the subject's "economic literacy" (Soper, 1979) and facility with manipulating dollar amounts: (1) "how much does the average American who owns a car spend on gasoline each month?"; and (2) if the price of gasoline increased by 10 cents per gallon, how much more would that person spend each month?" As with the risk experiment, we asked subjects to pause for a moment and think about how these four values of involuntary regulatory cost per annum might affect their lives: \$10, \$50, \$500, and \$5,000.

In both experiments, we then offered an "attention check" question (a list of colors to choose one's favorite among, but with a paragraph of verbiage instructing the subject to choose "both red and green" no matter what their true preference was.

4.3 *Definitions of De Minimis and "Intolerable" Risk and Cost:*

4.3.1 *Risk.* To carefully elicit a value for R_D (*de minimis* risk), we first asked subjects to consider very small probabilities with reference both to where a risk might be indistinguishable from zero (specifically, we asked subjects to estimate a probability "so small you would not spend a moment thinking about it"), but also with reference to the non-zero but vague benefits of the consumer product posing the fatal-disease risk (above the probability you are about to choose as R_D , you "might consider [risk] when you decide whether to use the product that brings the risk with it."). We felt this explanation effectively isolated *de minimis* risk as more of a practical/mathematical construct than exclusively an ethical one (we did not ask subjects, for example, to "choose a level below which you or anyone could be exposed to this risk without any disclosure or benefit"), but still recognizing that all choices involve tradeoffs and that "effectively zero" may depend to some extent on what one might receive in return. Before asking subjects to choose the denominator X as the *de minimis* level in a fraction $1/X$, we reminded them that "if you name a very small probability (that is, 1 in 'extremely large number'), you are in effect saying that almost any remote probability of grave harm is large enough to cause you to think about the benefits of this product, and to consider possibly giving it up for yourself, or accepting a ban on it for everyone, because of its risk. If you name a very large probability (1 in 'extremely small number'), you are instead saying, in effect, that to gain

the moderate benefits of this consumer product you are willing to consider almost any probability of harm as a level that wouldn't concern you."

To elicit "intolerable risk" (R_I), we emphasized that while *de minimis* effects are in part perceptual ("rounding down to zero" may not implicate ethical judgements), intolerability is largely a moral judgment. So, we specifically asked subjects to identify a probability of grave harm "at or above which you would strongly object, as an ethical matter, to ever having to face it." As before, though, we needed to caution subjects about offering doctrinaire answers that would either refuse to consider *any* risk level "tolerable," or that would accept substantial probability of harm out of concern that limiting risk would deprive the subject of important benefits. So we first defined R_I as a probability above which "you should never have to face a risk this large from a consumer product or other environmental source of involuntary exposure, even if you derive some benefit from it." We then provided a similarly more detailed instruction, grounding risk in the context of offsetting benefit, as we had done with R_D previously.²⁴

4.3.2 Cost. We defined *de minimis* cost (C_D) as "an amount of cost you might have to pay every year for controlling this cancer-causing pollutant that you consider to be effectively zero; a cost equal to or smaller than your chosen level is something you would not spend a moment thinking about." As before, we linked the cost to the concomitant benefit: "any larger amount and you might want to know why this cost was occurring; you might even try to oppose that increase in prices per year, even if keeping the prices unchanged would allow this medium-sized population cancer risk to persist as well." Because personal income may be an important determinant of how people perceive cost (see Section 5.8 below), we suggested that subjects should "feel free to think about either a cost that in fact IS so small relative to your income or wealth that you consider it trivial, or a cost that you might notice but feel that you SHOULD regard it as trivial, considering its purpose."

²⁴ Specifically, we told subjects that "if you name a very large probability (that is, 1 in 'extremely small number'), you are in effect saying that to gain the moderate benefits of this consumer product you are willing to consider almost any probability of harm as tolerable. If you name a very small probability (1 in 'extremely large number'), you are in effect saying that almost any probability of harm is too intolerable to justify the benefits of this consumer product, and you are willing to give it up for yourself and others (e.g., a ban)."

For “intolerable cost” (C_I), we instructed subjects to provide an annual level of regulatory cost that “you should never have to pay for controls on this cancer-causing pollutant, even if it is causing a medium-sized risk of cancer in the U.S. population; any cost less than that per year, and you would no longer be sure you would object. Feel free to think about either a cost that in fact IS so large relative to your income or wealth that you would find it impossible or a great hardship to pay, or a cost that you might easily bear but feel that it should not be imposed upon you despite its public-health purpose.”

4.4 Interindividual Variability in Perceptions.

For each of the goalposts in the risk experiment, we then asked subjects to provide two alternative answers, each representing their views of how other people might offer estimates of R_D or R_I that were higher or lower than their personal value. First, we asked for what amounts to a lower bound on R_D , considering the views of persons other than the subject who might be less willing to “round down to zero” than they were at their own *de minimis* response. We phrased this question as follows: “now choose a probability that is so small that you think *almost no one* would truly consider it to be of any concern.” Subsequently, we elicited the upper bound: “On the other hand, some people may have no concern about the small risk of 1 in [their original denominator] that you gave us earlier. So we also want to ask you to choose a probability that is small, but large enough that *almost everyone* would agree that it is at least of some concern.”

Subjects were then probed about the range of reasonable estimates by others of R_I : first a probability “so high that you think almost everyone would agree is intolerable,” and then a probability lower than (or equal to) their original answer that “almost no one would find intolerable.”

Because we presumed that personal income would have a strong effect on what amounts of cost subjects would consider *de minimis* and intolerable, and that asking subjects to speculate about the spread of others’ values for these benchmarks would amount to asking them about interindividual variability in income as much or more than in perception, we did not ask the (2x2) parallel questions about cost that we asked about risk.

4.5 Demographic and other Covariates Collected.

Opportunity samples were recruited through random sampling from the Prolific online panel, whose members receive small fees for answering surveys, of U.S. residents who were 18-plus years old and proficient in English; data collection occurred in December 2019 for the cost survey and February 2020 for the risk survey. The Prolific panel provided us with no personal data or IP addresses, and we collected no locational data except for confirming U.S. residence, so subject anonymity was preserved. In both experiments, we collected information on each subject's reported gender, age, highest level of education (one of seven categories), ethnicity, political party, ideology (7-point scale ranging from "extremely conservative" to "extremely liberal"), annual income (subjects could choose one of 34 categories, such as "\$100,000 to \$124,999"), and overall wealth, defined as all financial assets plus one's home (41 categories to choose among, including a category for debts in excess of assets). We also asked four questions to gauge each subject's comfort level working with numbers (subjective numeracy; Fagerlin et al. 2007), and then proceeded to assess objective numeracy as follows : (1) in the risk experiment, we asked four multiple-choice questions requiring some rudimentary knowledge of probability theory (e.g., "if 1 in 100 students qualifies to take a test, and 1 in 100 test-takers win a scholarship, what is the probability that a random student will win?"); and (2) in the cost experiment, we first asked subjects to estimate what marginal tax bracket they fall into (one further and very rough indication of "economic literacy"), and then posed five multiple-choice questions, three about probability taken directly from the risk experiment, and two financial questions (e.g., "If you had \$100 in a savings account with 2% annual interest, would you have more than, less than, or exactly \$110 after five years?").

5. Results:

5.1 Demographics and Risk Numeracy.

We compare sample and U.S. (mostly adult) demographics in Table 1, using 2020 American Community Survey 5-year estimates, as full 2020 U.S. Census data were unavailable at the time of writing. These show that our samples were similar in gender, non-Hispanic white ethnicity, and income to the U.S. adult population, but younger and better educated. Their political

tendencies were skewed towards the Democratic party and liberal political ideology, based on comparisons to Gallup polling data.

Table 1. Demographics of the Samples and the U.S. Population

Attribute	Risk Sample (February 2020)	Cost Sample (Dec. 2019)	U.S. Data
Gender (female)	49.9%	50.4%	51.3% (18+ years)
Age Median	31.0	33.0	38.2 (includes < 18)
Mean (s.d.)	34.4 (12.7)	35.6 (12.7)	
Education: high school or less	13.9%	13.8%	38.2% (25+ years)
Education: college graduate or more	57.1%	57.4%	32.9% (25+ years)
Ethnicity: Non-Hispanic white	72.9%	72.0%	70.4% (includes children)
Currently employed for pay, full-time	43.7%	50.2%	NA
Annual household income, median range and proportion	\$30,000-\$59,999 30.2%	\$30,000-\$59,999 31.5%	\$35,000-\$49,999 (15+ years old) 12.3%
Wealth (assets minus debts), median range and proportion	\$0-\$19,999 32.8%	\$0-\$19,999 20.4%	NA
Political partisanship, Democrat	49.0%	47.9%	NA
Political partisanship, Republican	15.1%	29.1%	NA
Political ideology, slightly to extremely liberal	64.1%	63.1%	NA
Political ideology, slightly to extremely conservative	17.6%	18.8%	NA

Note: US Census 2020 results were unavailable at the time of writing, so 5-year American Community Survey data for 2020 were used as a placeholder. NA = not available.

In the risk experiment, from our initial sample of 750 subjects, 78 of them either failed the attention check (told to select both red and green as their favorite color, regardless of their actual preference) or made an elementary mistake suggesting incomprehension of probabilities as fractions (e.g., their denominator X in their "one chance in X is trivial" was smaller than their denominator Y in "one chance in Y is intolerable"). In the cost experiment, 580 of the original sample of 600 subjects passed the attention check; 3 of the 580 (and 5 of the remaining 20) offered values for "intolerable cost" that were lower than their value for "*de minimis* cost," so the final sample for the cost study consisted of 577 (580 minus 3) subjects.

In the risk study, subjects were widely dispersed with respect to their ability to answer the four probability questions used to assess objective numeracy: 286 of them (43%) could only answer one or two of the four correctly, while 182 of them (27%) were able to answer either three or all four correctly. On a scale of 1 to 3 for each of six questions about familiarity with "everyday probability" (experience playing cards or dice games, knowing what "Russian Roulette" means), the median score was 16/18, evidencing broad familiarity across the sample. The sample for the cost study showed similar dispersion in the answers to the three probability questions carried over from the risk study: 363 subjects (63%) answered 0 or 1 question(s) correctly, while the remaining 37% answered 2 or all 3 correctly.

5.2 Cost and Economic Literacy.

Most subjects in the cost study answered the two "cost numeracy" questions correctly: 388 (67%) understood that at 2% annual interest, \$100 will grow to more than \$110 in 5 years, while 459 (80%) understood that if a savings account paid 1% interest while inflation was running at 2%, at the end of any time period subject would have less purchasing power than before. Subjects also did extremely well on an "economic literacy" question (see Section 4.2 above) that asked how much the average car-owner spends on gasoline each month (the correct answer, at the time of the survey, was very close to \$120/month, at an average consumption of 45 gal/month and the average price at the time of \$2.60/gal). The interquartile range of the 577 responses was \$80-\$200, and the median was exactly the \$120 amount that the U.S. car-owner spends on average. Subjects did not do nearly as well on the question about how much the average monthly gasoline bill would rise if the per-gallon price increased by 10 cents: given the

roughly 45 gal/month average, the answer to this question should have been about \$4.50, but the interquartile range here was from \$6 to \$103. These laypeople apparently have an excellent factual or intuitive grasp of how much the average person spends on a commodity like gasoline, but have difficulty manipulating numbers to estimate the effect of price changes. We took each subject's estimate of the monthly gasoline bill, divided it by \$2.60 to impute a number of gallons/month, and multiplied that number by the 10 cent fictional price hike; when we computed the ratio of the answer given (dollar increase in the monthly gas bill) to the logical answer given subject's own estimate of quantity, the interquartile range of that ratio was between 1.3 and 28. Many subjects, therefore, offered very reasonable answers for spending but grossly overestimated the effect of a 10¢/gal increase on that spending.

5.3 Completion Time.

Despite the lengthy instrument and the emphasis on questions requiring careful and quantitative thought, subjects were able to complete the experiment in a reasonable period of time. For the risk survey, the median time of completion was 26 minutes, with an interquartile range of [19 – 34 minutes]. For the cost survey, the median was 21 minutes, with an interquartile range of [16 – 28 minutes].

5.4. De Minimis Risk.

The function colored in black in Figure 2 shows the distribution of the 672 responses to the question “what probability of grave harm is so small that you would not spend a moment thinking about it?” *Both the central tendency and the dispersion of these responses provide a fundamental set of new information valuable for risk perception and management, based on careful elicitation of subjects provided extensive information about probability and risk.* The best single estimate of *de minimis* risk, we believe, is the geometric mean of these responses, which is $10^{-5.89}$, or 1 chance in 770,121. This estimate is, obviously, close to but slightly higher than the canonical value of 10^{-6} (1 in one million) that has often been used for *de minimis* risk. But this divergence is not particularly significant, given that 28% of all respondents gave exactly 10^{-6} as their answer.

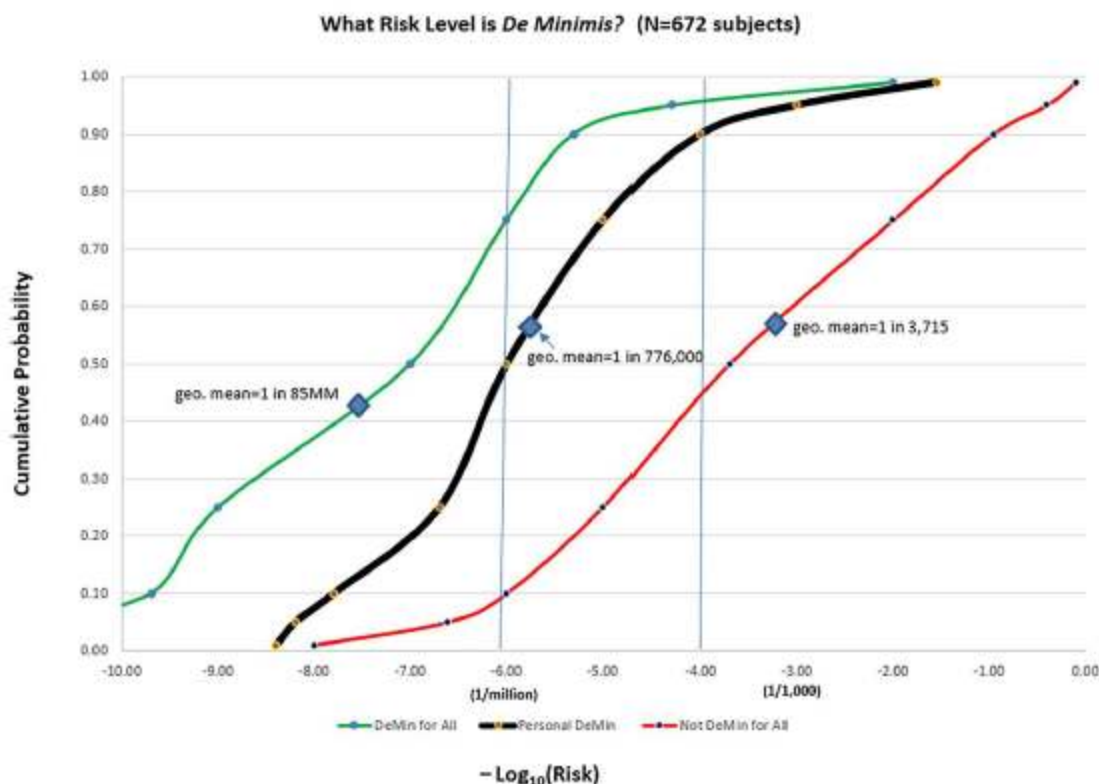


Figure 2. Cumulative distributions of *de minimis* risk responses.

Perhaps considering the dispersion of these values will be more useful for risk management. We can summarize this as follows: on the one hand, 61% of subjects (from the 16th to the 77th percentiles of the distribution) believe that a value within a factor of 10 in either direction from 10^{-6} is where *de minimis* risk falls—suggesting that the folklore about 1/million has been reasonable. On the other hand, 11% of subjects believe that a risk must be *smaller than 1 in 100 million* for it to be “too small to think about,” while about 10% of subjects are willing to ignore risks at 1 in 10,000 or higher.

If we consider the additional variability revealed when subjects were asked to imagine upper and lower bounds for *de minimis* risk across our diverse population, the spectrum of reasonable estimates of R_D widens significantly, as the green and red curves in Figure 2 show. However, it is arguably not reasonable to incorporate the upper and lower bounds on a large set of responses that themselves represent bounds. If instead we look only at the geometric means of each of the three distributions, the range extends from 1 in 27 million ($10^{-7.43}$) to 1 in 3,983 ($10^{-3.6}$). These endpoints represent roughly the 6th to the 85th percentiles of the responses to the

original *de minimis* question, but are still sufficiently wide-ranging that risk management policy would find it difficult to accommodate all these views with a single conception of *de minimis* risk.

Having reported this first main result, Figure 3 below now depicts its interquartile range in context of selected prior estimates of R_D that were not based on direct elicitation from citizens:

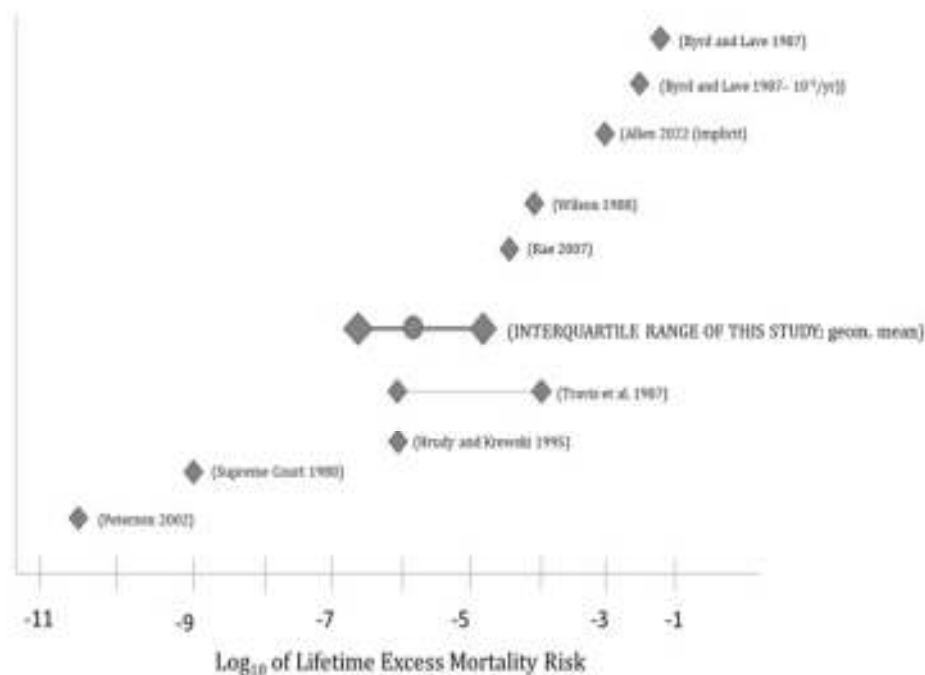


Figure 3: Prior Estimates of *De Minimis* Risk, Including this Study's Result

This Figure suggests that our central estimate of 1/770,000 is a *more strict* criterion for a risk that can be ignored than many prior answers to this question have yielded.

5.5 Intolerable Risk.

The responses here were of particular interest, because unlike the case for *de minimis* risk, there is no “standard value,” not even a folklore value like 1/million.

Figure 4 shows the distribution (black curve) of 672 responses to “what is the probability of grave harm at or above which you would strongly object, as an ethical matter, to ever having to face it?” The interquartile range of this set of responses went from 1/10,000 to 1/20; 79 subjects (12%) offered exactly 1/1000 as their answer, and 108 subjects (16%; the modal answer) offered exactly 1/100.

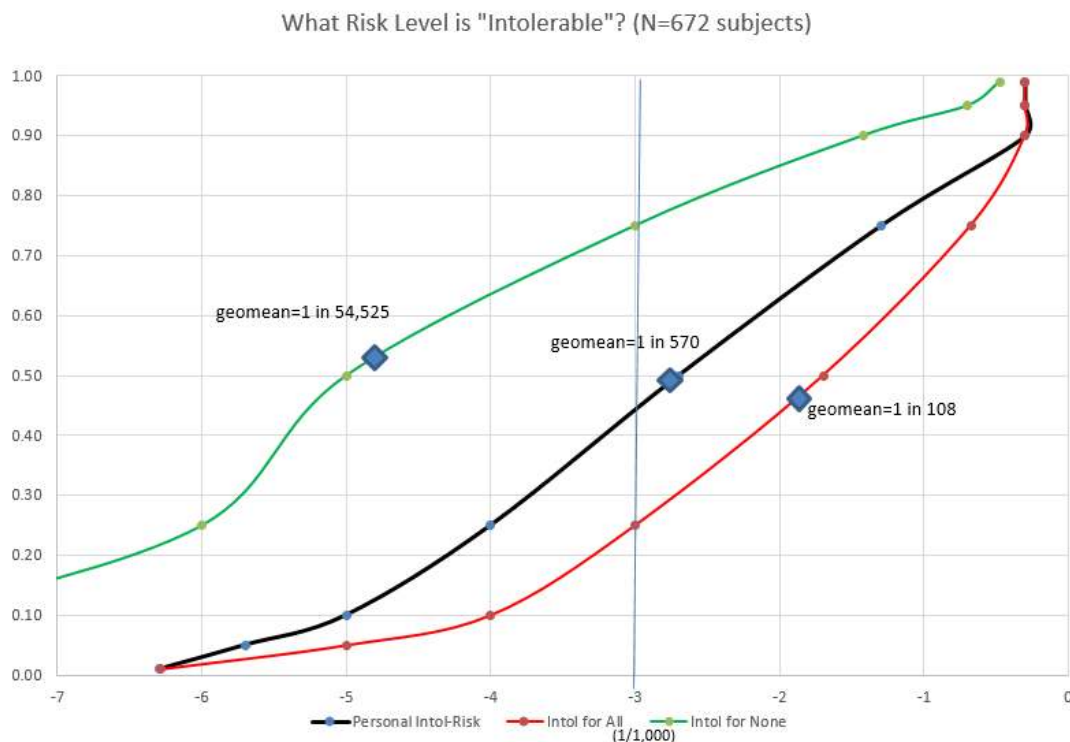


Figure 4: Cumulative Distributions of Responses to “Intolerable Risk” Question

As with the *de minimis* responses, the geometric mean and the median of the inter-subject distribution were quite similar: here the GM was a risk of **1 in 570**, and the median was a risk of **1 in 500**.

Both of these central-tendency measures of “intolerable risk” are only slightly higher than the OSHA level of 1/1000 that allows the federal government to “declare victory” and cease reducing risks in the workplace. This result, and the substantial fraction of respondents (25%) who view a 1/10,000 risk as intolerable, *suggests that 1/1000 is not low enough to be content with, in the workplace or in the general environment.* In other words, OSHA’s 1/1000 stopping point misreads the *Benzene* decision; this risk level is barely acceptable, and so by definition slightly smaller risks cannot be “insignificant.” And for EPA, whose risk-based goals include 10^{-4} (in, e.g., the Superfund program) and 10^{-6} (in, e.g., the Clean Air Act), *the former definition for “acceptable” may in fact be a risk level that one-quarter of its constituent citizens in fact view as intolerable.* At the very least, the central tendency of this distribution (roughly 1 in 500) reveals that there may be a very narrow window (about a factor of 2) between risks high enough to be flatly intolerable to most citizens, and those low enough to be accepted by some as regulatory stopping points.

The additional variance contributed by respondents’ impressions of what level all, or none, of their fellow citizens would find intolerable can further illuminate public choices about which risks must be reduced. That said, the variation within the outer curves in Figure 4 may, like the case with the *de minimis* responses above, allow for more “wiggle room” than is sensible. Concentrating on the geometric means of the green and red curves, we can see wide agreement that no one should flatly object to risks below about 1/50,000 (some risks of this magnitude may well have benefits that amply justify them), but that no one should ever be content allowing involuntary risks in the 1/100 range to persist, regardless of their offsetting benefits. This last conclusion does not hold, of course, if the large risk cannot be reduced by any feasible means; this is just an upper bound on the approximate risk level that all seem to agree can never be excused as “tolerable.”

Additionally, we note that all respondents seemed to understand that R_I must be a larger probability than R_D , which requires the answer given for R_I to be larger than for R_D (again, subjects were asked to give the answer X to questions of the form “what risk of $1/X$ is *de minimis* or intolerable?”). However, some subjects offered values for R_I and R_D that were either both rather large or rather small; for 15% of the sample, the two goalposts were within a factor of

10 of each other. The interquartile range for the full sample (ratio of R_I to R_D) went from a factor of 20 to a factor of 10,000.

5.6 *De Minimis Cost.*

As seen in Figure 5, the distribution of responses as to what annual involuntary cost respondents consider “effectively zero” had a central tendency of \$25 (both median and geometric mean had this same value) and an interquartile range of from \$10 to \$100. The most common single answer was \$10. We emphasize how much less variance there was around *de minimis* cost as opposed to *de minimis* risk: the ratio of the 90th percentile to the 10th percentile of the distribution was nearly a factor of 10,000 with respect to risk, but only a factor of 83 (\$250 versus \$3) with respect to cost. We also note that at a 3% discount rate, \$25/yr. represents approximately \$725 as a lifetime (70-year stream of payments) cost, a finding we will return to later.

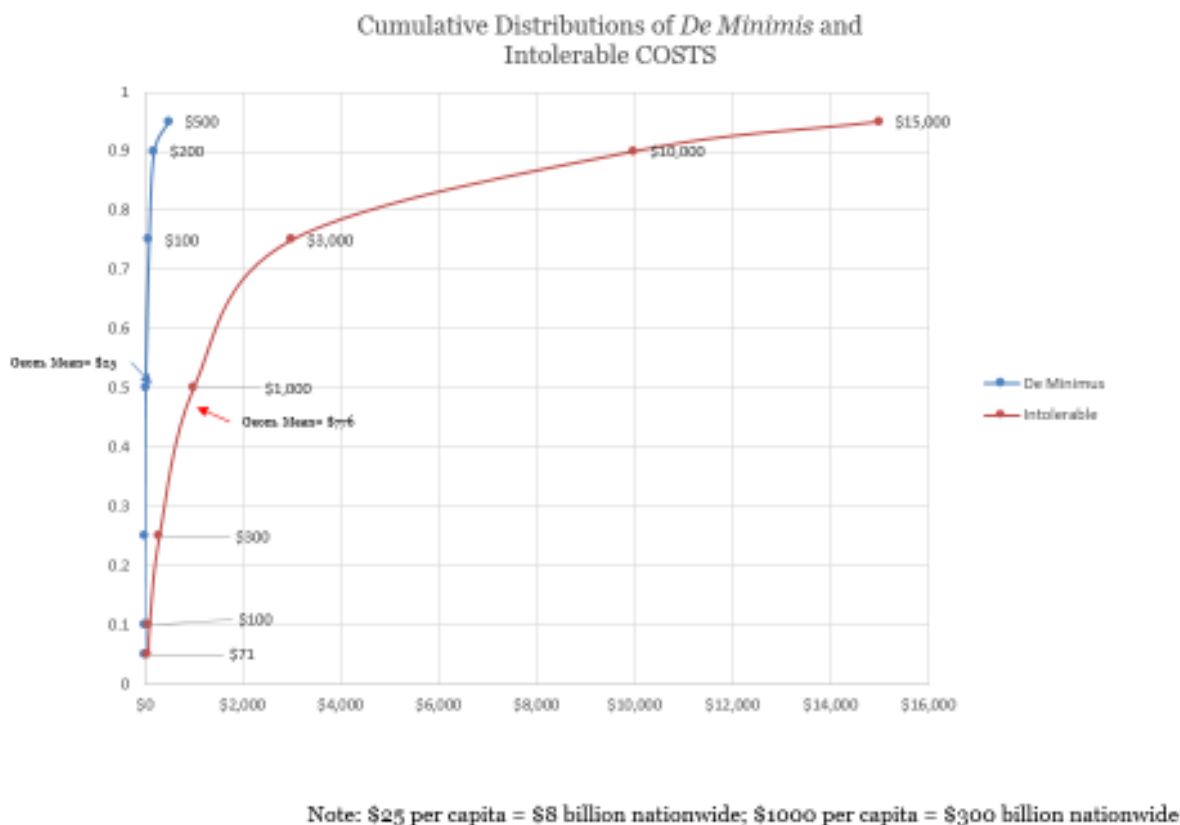


Figure 5: Cumulative Distributions for both Cost “Goalposts”

We know of no prior attempt(s) to define “*de minimis* cost,” either by imposing expert preferences upon the populace or by eliciting public views. However, one relevant benchmark here is the long-standing White House (OIRA) policy of requiring fuller review only of “major” regulations, defined as incurring more than \$100 million in annual cost to the economy. For regulations with very broad impact, this amount might represent as little as \$1 per household per year—an amount our subjects said was *far* below any amount they would consider “no longer equivalent to zero.” *So perhaps the notion of what regulations are “major” needs to be ratcheted upwards.* At our median response of \$25, a regulation that affected every household equally would impose a \$2.5 *billion* annual cost on the economy, and yet would still barely affect any household in any tangible way, according to this set of responses.

We also used the information collected on each subject’s annual income to convert her response for *de minimis* cost from absolute dollars into “weeks of salary.” As Figure 6 shows, the geometric mean response represented 1.2 *hours* of work at each respondent’s respective salary, with an interquartile range of from roughly 1/3 to 3 hours work (0.008 to 0.08 weeks).

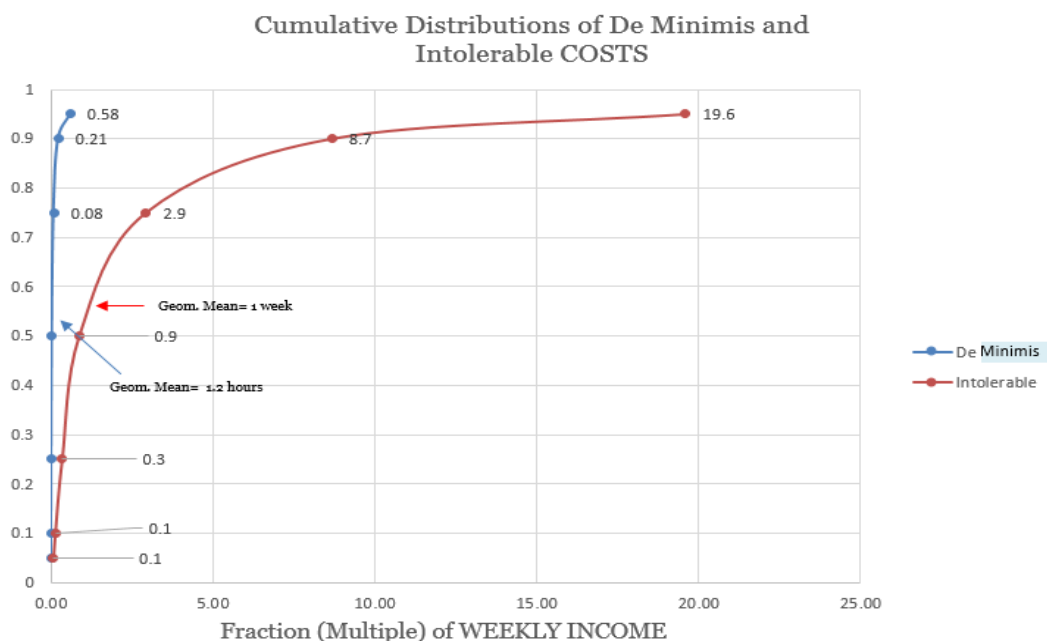


Figure 6: Cost “Goalposts” as Functions of Subjects’ Weekly Income

5.7 Intolerable Cost.

The median response for “what amount of annual regulatory cost you are sure you would object to anyone having to pay” was \$800, very close to the geometric mean of \$776. The interquartile range was from \$300 to \$3000, the same (rough) factor of 10 that offers a fairly robust consensus on what magnitude of cost is either intolerable or *de minimis*. When expressed as a fraction of weekly income, the geometric mean of these responses was almost exactly one week’s salary, with an interquartile range of from roughly 1/3 to three weeks. Thus, our subjects have offered two easy-to-remember benchmarks for these fundamental but rarely-discussed perceptual breakpoints in how laypeople view their monetary contribution to a worthwhile social/regulatory program; *devoting one hour per year of labor to the social purpose is seen as approximately “too little to consider,” while devoting one week per year is seen as a major, perhaps an impermissible, infringement.*

So, the ratio of intolerable to trivial is about 40:1 when cost is the subject (1 week’s to 1 hour’s labor), whereas for risk, the ratio is about 1350:1 (1 chance in 770,000 versus 1 chance in 570).

5.8 (Little if any) Effect of Income on Variance of Responses.

Interestingly, the dispersion of responses, for both C_D and C_I , barely decreased when we converted each response from absolute dollars to [fraction of weekly income]. The ratio of the 75th to 25th percentiles of C_D was 10:1 when expressed in dollars, and also 10:1 when converted to fractional income (the 90th:10th ratios were also similar, at 67:1 and 70:1 respectively). For C_I , the 75th:25th ratios were also virtually unchanged (10:1 versus 8:1), although there was somewhat less dispersion when the 90th and 10th percentiles were compared (100:1 for dollars; 56:1 for fractional income).

If laypeople tend to construe the amounts at which costs become trivial or intolerable in terms of their own income or wealth, then the dispersion of responses *should* markedly decrease when their dollar responses are normalized to fractional income. Since the dispersion of the absolute-dollar responses was not generally any larger than the dispersion of the “weeks of income” responses, it seems that *laypeople may tend to regard costs in absolute terms*, looking beyond their own prospects to view C_D and C_I as ethical benchmarks—amounts that no one

should care about or amounts that no one *should* have to pay, as opposed to amounts that each subject personally cares about or what s/he personally cannot pay.

5.9 Associations between Covariates and Responses.

In examining the effects of a large number of possible covariates on subjects' four main responses (D and I, for both risk and cost), some interesting patterns emerged. In general, there were no significant effects of age, sex, or educational attainment on subjects' tolerance for very small risks (and costs) or intolerance for very large ones. This need not be construed as a "negative" finding, though, because the lack of significant effects implies that the distribution of responses for the four outcome variables reported above may apply to U.S. citizens in general, and that decision-makers would not need to adjust them if trying to gauge the risk and cost perceptions of any major subgroup of the population.

However, we did find significant effects of objective numeracy: subjects who answered three or four of the questions correctly tended to demand significantly *smaller* risks to reach levels they considered *de minimis* than those who only answered one or zero questions correctly. The geometric mean of R_D values for the most numerate group was 1 in 945,400 ($10^{-5.98}$), whereas for the least numerate group it was 1 in 602,000 ($10^{-5.78}$). At the same time, the most numerate subjects regarded substantially lower risks (1 chance in 1,500, on average) as intolerable than the least numerate subjects did (1 chance in 250 for these respondents). *It therefore appears that more numerate subjects require risks to be lower for them to be not-intolerable or to be of no concern.* For both comparisons (low vs. high numeracy), the p -value for the difference of geometric means was < 0.0001 .

This finding represents a special case of a more general pattern, with a high correlation ($r = 0.55$, $p < 0.0005$) between subjects who demanded lower values of D also demanding lower values of I. In other words, people who are very concerned about small risks (refusing to declare something D unless it is very small) *also* tend to be the ones who find the relatively smallest risks intolerable. For example, the 161 subjects who considered risks 10^{-7} or below to mark the boundary of *de minimis* considered risks of 2.7×10^{-4} (geometric mean response) to be intolerable, while the 87 subjects who considered risks of 10^{-4} or above to be *de minimis* were comfortable

with risks as high as 4.8×10^{-2} (1 chance in 21) to be barely tolerable. So, there clearly exist population subgroups (the more numerate, and to a lesser extent the most politically liberal) who are uncomfortable regarding 1/million risks as truly *de minimis* and at the same time are uncomfortable with even those “small” risks that fall below the Supreme Court OSHA benchmark of 1/1000, probabilities many of them regard as “intolerable under any circumstances.”

6. Subjects’ Explanations for their Responses:

6.1 *De Minimis Risk.*

We asked subjects to indicate which information, if any, was most helpful that we provided to ground their responses in context of familiar probabilities, and then asked specifically (open-ended text responses) for insight into their thought processes as they were contemplating each question.

When asked about R_D , subjects most often cited actuarial examples of risks in daily life (all 672 subjects answered this question, and 78 of them mentioned actuarial information) as the most helpful info, followed by the Russian Roulette images (46), odds of rolling dice (33), icon arrays (26) and odds of poker hands (20). The probability of death in a “bouncy castle” (1 chance in 4 million per lifetime) was by far the most frequently-mentioned benchmark that helped ground the answer for D .

There were several recurrent themes in how subjects arrived at their estimates of D . Most often, subjects explained their thought process using this logic: they thought first about a denominator that reflected their “universe” (many chose the US population, some the world population, and others their city or state, while still others thought about a large crowd or a stadium full of people), and then made a subjective judgment about the numerator (how many deaths would be acceptable among this group?). This resulted in a variety of R_D estimates—for example, someone who said “330 deaths from this cause would be too many in the US” would tend to require an R_D value below 10^{-6} (330/US being 1/MM). Here is a representative quote from one subject who used this mental model:

“I imagined the global population, rounded to 8 billion just for ease. I imagined a 1 in million chance as affecting 8000 on the entire earth, which is more than effectively zero. One in 500,000, affecting 16,000 people on earth, is still zero to me. One in 100,000, affecting 80,000, is starting to appear more than zero.”

Many other respondents arrived at values around 10^{-6} for R_D by asserting that one excess death is acceptable in a [denominator such as a large city], but more than that is too many. There were many permutations of this mental model, affecting both numerator and denominator, but this two-stage thinking was very common.

In contrast, many other respondents focused on an actuarial risk we provided that they “had never realized existed” or “knew it could happen but never worried about,” and used that probability to analogize that it must be trivial in general. The 1 in 4 million risk of dying in a “bouncy castle” came up most often here, and some explicitly used two probabilities (e.g., deaths from dog bites versus bouncy castles), one of which they regarded as non-trivial and the other as clearly trivial, to choose as R_D a value in between the two.²⁵ A few subjects mentioned actuarial risks that they had personal knowledge of, such as a 10^{-4} risk of getting epilepsy from watching video games, as something that “must be” *de minimis* because they take the risk themselves without thinking about it.

Finally, there were significant numbers of subjects who offered that the phrase “you’re one in a million” is salient and must have some relationship to a frequency we are wired to regard as the lower limit of possible/worthy. There were clearly a few protest responses, some who picked 1/100 million when our system prompted them that risks at 1/billion or less were unreasonably low, and several “life is infinitely precious and no risk is trivial” responses.²⁶

Table 2 provides some idiosyncratic and perhaps interesting responses, some of which offer different thought processes than those described above.

²⁵ Similarly, a few subjects used “something positive that’s never happened to them or anyone they know,” like getting the grand prize in a lottery, to estimate D by analogy.

²⁶ We conducted the survey at the very leading edge (March 1, 2020) of the Covid crisis, and a few subjects cited the Covid death rate of 2.3% (although this was at the time the case fatality rate, not a risk) as helpful to their answer.

TABLE 2
Idiosyncratic Open-Ended Responses to *De Minimis* Risk Elicitation

Personal Analogies:

- ($R_D = 1/500,000$) “I thought back to medications where I experienced side effects that were supposed to be very rare.”
- ($R_D = 1/1,500,000$) “I have had a life where seemingly impossible things have happened to me (being molested, electrocuted, had a child with a birth defect...) and this seems to be a low enough chance to make me feel better.”
- ($R_D = 1/10,000$) “1 in 10,000 is how unusual it is for someone to have situs inversus (my Dad has it, so that's why I know the statistic) and I don't really think about how frequently the people I meet might be like my Dad, due to the rarity.” [note: Situs inversus is a benign genetic condition in which all the internal organs in the chest and abdomen develop in a mirror-image (left to right) position to the “normal” arrangement.]

Salience of Benefits (despite our emphasizing that the hypothetical consumer product is nonessential):

- ($R_D = 1/80,000$) “Using a lawn mower at least occasionally is somewhat essential to our way of life. I had to weigh the pros and cons of the situation to come up with how essential that activity is.”

Non-Risk Analogies:

- ($R_D = 1/1,00,000$) “I thought about 1 in 1,000,000 being basically the chances of winning the lottery, and I thought about how huge \$1,000,000 is and how small 1 in 1,000,000 is.”
- ($R_D = 1/100$) “I picked 1/100 because I was thinking about school grades. Essentially getting one question right on a 100 question quiz is a zero.”
- ($R_D = 1/1,000,000$) “Sometimes people say ‘you are one in a million’ but they never say it about me, so it seems pretty safe.”

6.2 Intolerable Risk.

When asked to explain their choices for the magnitude of R_I , fewer respondents ($N = 25$, versus 78 for R_D) mentioned actuarial risks as helpful in grounding their responses; of these, the most often-cited benchmark was the odds of a fatal dog bite (1 in 133,000), which 10 subjects used to justify rather stringent choices for R_I (presumably expressing the view that anything more common than a fatal dog bite is “intolerable”). The proportion of subjects citing daily-life probabilities (card games, dice) as influential in their thinking was similar to that for the R_D responses.

Many subjects used the same numerator/denominator mental model we saw in the previous question, in this case most often choosing a denominator *small enough* such that even one grave event in the numerator would be “intolerable” to the respondent.²⁷ By this logic, the number of one’s Facebook friends, or her graduating class, or “double the size of my family,” led to a choice for R_I that was the reciprocal of that sized group. A variation on this thinking led a few subjects to conclude that (for example) because one fatality in a stadium full of people could not possibly be “intolerable,” the magnitude of R_I would have to be substantially larger than this reciprocal. In some contrast to the elicitation of R_D , several respondents specifically mentioned the area of drug safety as very helpful in grounding what is intolerable; for example, one respondent opined that a medication that has more than a 1% chance of causing a grave side-effect is “intolerably dangerous,” and thus offered 1×10^{-2} as their response for R_I .

6.3 De Minimis Cost.

At first glance, one might not expect much of interest to emerge from subjects’ explanations of what amount of involuntary (regulatory) cost is “too small to give thought to.” Indeed, most subjects did report some variation on the simple mental process of “what do I spend money on such that I don’t miss the money or could easily do without the purchase?” Various respondents mentioned “one extra cup of coffee per week/month” as a guide to estimating *de minimis* cost, while various others honed in on a purchase (e.g., an expensive pair of headphones) that they

²⁷ Not every respondent imagined mortality in the numerator; one subject chose $1/35$ as intolerable because of the probability of “being called on in class when you don’t know the answer.”

regarded as *not* trivial, and then chose C_D as some amount below that benchmark (a few brought up opportunity costs, by mentioning that an amount not *de minimis* is by definition an amount that causes them to defer some *other* purchase due to its size). Many subjects mentioned that the table we provided (giving total and per capita annual expenditures on different public programs and consumer goods) helped them arrive at an answer; 21 subjects specifically mentioned the \$25 per capita spending on potato chips as helpful, while 14 mentioned the \$6 per capita expenditure to effectuate the roughly 1000 heart transplant operations in the U.S. each year.

But there were some interesting insights within the open-ended individual explanations of where each C_D value came from:

- With the recent emergence of monthly auto-billed subscriptions to streaming and other services, many subjects mentioned such amounts (roughly \$12/month) as amounts that they “don’t even remember they are being dunned,” and hence evidence that they regard such sums as *de minimis*;
- A few subjects sought an amount of *time* they regard as *de minimis*, and reported the dollar amount, as in “one day’s salary per year is *de minimis*”;
- Some respondents calibrated their answer to an amount that is not necessarily trivial, but that they are required to pay, such as their health insurance premiums or their share of the national defense budget. So for some, “not worth a second thought” may relate to expenditures where the “second thought” is fruitless rather than unnecessary;
- Perhaps most interestingly, given the prior suggestions in the literature on *de minimis* risk (see Section 3.1 above) that thinking about perturbations may be more useful than baseline amounts, some respondents explained that they estimated their C_D value by considering *changes* in the price of some commodity (most often, gasoline) that are small enough to be imperceptible against the background expense they are used to bearing.
- Finally, a few subjects explicitly thought about what amount would be *de minimis* for the entire nation, and offered their per capita share of that amount as by definition *de minimis* for them (e.g., “\$50 is C_D because \$15 billion is rounding error in the federal budget”).

6.4 Intolerable Cost.

There were two closely-related themes in many of the comments from subjects explaining why they chose a particular value for C_I : an “intolerable” regulatory cost was equal to or greater than an amount they were already spending for something that seemed more important than risk reduction, or than an additional expenditure that, if required, would present “intolerable” financial hardships. Examples of the former theme included “half of my annual salary,” or “one month’s rent,” or “one year’s tuition”—all expenditures that helped these subjects gauge what amount of additional cost would be out of proportion for anyone to have to spend for a policy that would eliminate a “medium-sized cause of death” for the nation. Other subjects, though, gave examples such as “enough expenditure to make me homeless,” or “more than I could afford even if my life depended on paying it.” Some respondents specifically mentioned an amount that would induce a particular negative emotion, such as “the expense for a previous major surgery that I thought at the time was unfair,” or “an amount I would resent having to pay,” or “an amount that would put a crimp on our ability to save and to enjoy life.”

In addition, some respondents explicitly started from an amount they judged would be intolerable (or unrealistic) for the nation as a whole, and then divided that number by 300 million to arrive at an estimate of C_I for an individual. Others used the obverse logic, and increased their estimate of C_I to a point at which 300 million times that number would become unrealistic for the nation, or would exceed some benchmark like the national defense budget that the subject felt was clearly more important than eliminating a “medium-sized cause of death.”

Only a few subjects explained that they took into account the ability of the poor to bear a cost that they personally might be able to tolerate. Although in this experiment we asked subjects to think only about their personal situation (see the exact language for this prompt in Section 4.3.2 above), a few respondents reported that they had lowered their response to C_I to take account of differential ability to pay.

7. Limitations:

Our sample was an opportunity sample not necessarily representative of the U.S. adult population. The sample exhibited similar gender, non-Hispanic white ethnicity, and income to

population-wide estimates available from the U.S. Bureau of the Census, and multivariate analysis suggested that none of the four risk or cost boundaries differed significantly across demographic categories. However, the sample was younger and better educated than U.S. adults generally, and a single study cannot be used to definitively reject the hypothesis that *de minimis* and intolerable boundaries are associated with different demographic sub-groups. We also stress that extrapolation to non-U.S. populations is unwarranted, as we cannot assume that their responses would be identical even when pegged to comparable measures (e.g., proportion of income for cost responses).

As noted earlier (footnote 22), we designed the experiment to focus on a vaguely-defined hazard, so as to avoid idiosyncratic and non-generalizable responses. However, we suggested that the hazard *might* be thought of as something like a “polluting lawnmower,” to be consistent with social science experience generally (e.g., not just limited to CBA) that eliciting evaluations from citizens yields more valid and reliable results if the item to be evaluated is narrowly described. This analogy may have failed, for some subjects, to elicit perceptions about a hazard whose concomitant benefits were neither trivial nor irreplaceable (see footnote 21). We therefore believe it would be useful to replicate this work using different products, hazards, and exposures to estimate the robustness of the results reported here to these variations.

We took care to elicit these boundary values in ways that would minimize biasing citizen responses. However, the CBA literature, among other fields, has developed a wide range of frequency statements, risk ladders, or grids (Cropper et al., 2011) as decision aids to help people grapple with small, unfamiliar probabilities, and one task of studies building upon our initial findings would be to probe the degree to which varying information-provision might affect results (as we have done with respect to our series of experiments with “macro-risk tradeoffs,” Johnson and Finkel 2022a). Although we expected that assessing boundary values for *de minimis* and intolerable costs would be less problematic, our observation that (for example) some people seemed to take others’ tolerance into account instead of focusing on their own tolerance, and the general principle of treating benefits and costs comparably in CBA, suggest that testing the variability of cost boundary elicitation across methods also would be advisable.

Although they are not technically “limitations” of the study overall, there are two other caveats to mention. First, we did not probe (beyond optional open-ended responses) the basis on which people selected a given boundary as *de minimis* or intolerable, a question requiring further detailed planning and implementation. For example, our data do not allow us to determine whether the choice of 1 in one million as the R_D stems from this round number being culturally available (e.g., “you’re one in a million!”) in the U.S. population, or specifically learned in college or other classes and parroted back to us here. With regard to the latter potential explanation, more people choosing exactly 1 in one million as their *de minimis* risk had a college degree (67.8%; $n = 87$) than those who did not select that *de minimis* value (55.7%; $n = 663$; $t = 2.411$, one-sided $p = .009$). But we cannot conclude that it was school learning about expert-proposed *de minimis* boundaries that increased the choice of this answer without knowing more (e.g., about the content of classes taken).

Secondly, for space reasons we reported results that give us the ability to distinguish *de minimis*, intolerable, and “in between” harms to health or pocketbook, but have not yet reported information about how laypeople might quantify their values for those risks or costs in the large mid-region. As noted earlier, in the experiments discussed here we also led subjects through a structured elicitation of values spaced evenly between their personal “goalposts.” In a paper in preparation, we will report those results, combining them with this article’s results so as to offer a principled way to apply an empirical social welfare function to any interindividual distribution of risks and costs, thereby estimating the sum of welfare changes to every individual.

8. Discussion: Implications for Policy

8.1 Which Small Risks Should be Ignored?

The central curve in Figure 2 offers two somewhat divergent lessons for those concerned with appropriately reacting to *de minimis* risks, depending on whether we focus on the central tendency of the responses or the substantial interindividual variation in the perception of *de minimis* risks, and further on whether we choose to interpret the latter in a precautionary way.

The geometric-mean response of 1 chance in 770,121 suggests that the canonical value of 1/million for *de minimis* risk might need no revision, or need only a slightly more tolerant risk management view, one that regards some risks slightly higher than 1/million as also unworthy of any attention. The observation that 61% of respondents were comfortable with a dividing line between *de minimis* and *de manifestis* risk somewhere between 10^{-5} and 10^{-7} adds support for the view that, however *ad hoc* and unsupported by any empirical evidence the initial “one in a million” rule of thumb was decades ago, it was a wise or at least a lucky choice (unless, somehow, significant numbers of respondents knew of 10^{-6} as the common regulatory goal and were influenced to repeat that value as their own).

*This result is particularly salient given the amount of scorn heaped upon 10^{-6} over the decades. Yes, this increment is very small compared to the roughly one-in-three chance of developing cancer in any one person’s lifetime, as Byrd and Lave (among many others) have forcefully stated: “An increase of 1 cancer in 1 million lifetimes means that the incidence of cancer will be 300,001 instead of 300,000... We would instruct regulatory agencies to ignore such trivial risks.” But to the extent that public agencies should only “ignore” risks that the public actually wishes them to ignore, the results here should restrain any advocacy for raising the *de minimis* threshold closer to 10^{-5} or higher.*

Indeed, these results also indicate substantial support for lowering the 10^{-6} dividing line, as more than ten percent of all subjects, having carefully considered probability and severity, chose a value below 10^{-7} as the lowest involuntary (grave) risk they would worry about and would welcome public intervention to reduce. Continuing to ignore risks below 1/million would disenfranchise this substantial fraction of the U.S. population, who perceive 10^{-7} (or even lower) as palpable. Marginalizing ten percent of the population is clearly not a “reasonable worst case,” or as inclusive of uncertainty or variability as the traditional use of $p < 0.05$ is as an expression of how strongly we want to guard against over-reaction to risk. On the other hand, an openly “anti-precautionary” value judgment (in essence, expressing precaution against over-regulation) might relax the *de minimis* criterion to 10^{-5} or even higher, ignoring risks below that size, on the grounds that nearly one-quarter of all respondents were blasé about risks at these levels.

When we broaden our sights to look at the levels of risk subjects thought that “almost no one would consider” to be other than trivial, versus the higher levels that subjects agreed “almost no one would consider to be *de minimis*,” we can offer additional support *either* for a tightening or a relaxation of which risks are small enough to ignore, again depending entirely on whether we want to be conservative or not. However, here one might argue that some double-counting exists, to the extent that the most risk-averse among us may be imputing extreme risk aversion to hypothetical others (and conversely, that the most risk-tolerant are relaxing the *de minimis* criterion by imagining still others who are even more blasé than they are). The fact that the range from the lower bound of the green curve in Figure 2 to the upper bound of the red curve encompasses nearly 8 orders of magnitude suggests that the “what do you think others think?” cumulative distribution functions are of limited use for public policy.

We emphasize that whether federal and state agencies change or reinforce the particular 10^{-6} benchmark for *de minimis* risk, *the time has come to reify the de minimis concept*. If we truly believe that some individual risks are tantamount to zero, no matter how many “statistical lives” might be lost within a massive population facing such risks, then we need to take two actions in response:

- We would need to enshrine the stopping point in law or practice: no regulation should be made so stringent that not even the most at-risk individuals would face risks above the *de minimis* level.
- We would need to truncate the way we convert risk distributions into measures of total benefit: no individual risks below the *de minimis* cutoff should be part of the summation (integral) of [risk times number of persons at that risk level] that results in a “statistical lives saved” calculation.

To fail to make these policy changes would be to continue to regard the *de minimis* concept as something deserving only of lip service: instead, *risks that are tantamount to zero should not be reduced further* (and we hasten to point out that the next subsection of this paper offers a countervailing prescription).

8.2 Which Risks are so Large that their Reduction Should be Valued at More than the “Fractional Lives Saved”?

Figure 4 reveals that a substantial minority of respondents believe that society should eliminate all involuntary risks that are as low as 10^{-5} , which is clearly an unrealistic and counter-productive hope. At the other extreme, some subjects are unwilling to rule out involuntary risks as high as 1 chance in 2 as being “tolerable,” and clearly such risk tolerance cannot sensibly be applied to the population at large.

But the central tendency of this distribution may be quite useful and revealing: the GM of these responses was a risk of 1 in 570, and more than 55% of respondents regard a risk of 1/1000 as intolerable. *This suggests that in many cases where public policies reduce average risks to roughly 1/10,000—especially OSHA policies that reduce only those risks that are detected and properly abated, and then only to 1/1000—some or many citizens are left bearing risks that the majority of us believe are in fact not tolerable, fair, or reasonable.*

Society should therefore consider the same two implications of intolerable risk as we discussed above for *de minimis* risk:

- We would need to [try to] enshrine this stopping point in law or practice: no hazard should be left unregulated, or regulated so weakly, that significant numbers of citizens would face risks above about 1 in 500.
- We would need to alter the way we convert risk distributions into measures of total benefit: individual risks near the level at which they become intolerable should **not** be treated the way we treat smaller risks, as merely a part of the summation (integral) of [risk times number of persons at that risk level] that results in a “statistical lives saved” calculation.

This second point deserves elaboration, as unlike the analogous recommendation above (to regard all risks lower than *de minimis* as of zero weight), it will often not be possible or desirable to give intolerably high individual risks infinite or even enormous weight. *But this fact is no excuse for treating intolerably high risks as merely “more fractional lives.”* The glaring problem here is that for decades, economists and others who have done so much to advance the way we value life-prolonging benefits have often cautioned that the estimates used for the VSL only apply when the risks are “small” and should not be used for very high risks, as doing so would under-value them. For example, Viscusi (2007, p. 114) notes: “[T]he VSL will understate

how much people must be compensated to face a series of increases in fatality risk that culminate with certain death.” But that important caveat always seems to end with the aside or footnote, and is never carried forward into regulatory policy or risk communication; *to date, no one has ever suggested at what magnitude an individual risk ceases to be “small” for this purpose, nor any principled, quantitative way to give “not small” risks greater but not infinite weight.* As a result, we have tacitly NOT given them greater weight, which is a highly value-laden choice.

Space limitations here preclude the presentation of findings from the parallel study we conducted to the one reported here, but a manuscript in preparation will discuss how we elicited estimates from laypeople about the *shape of their disutility function over risk (and over cost) as risk (or cost) increases from the level each subject regarded as de minimis to the level s/he regarded as intolerable.* The parameterization of these functions, averaged over the population, could eventually prove useful in converting “aggregate lives saved” benefit measures, and “total dollars” cost measures, to the more theoretically-correct measures of “aggregate changes in individual welfare offered by risk reductions.”

Together, the results presented here from the two risk elicitations suggest that laypeople on average divide risks into three categories: (1) risks smaller than about 10^{-6} , which most people are comfortable not receiving social attention; (2) risks larger than about 1/1000, which most people insist be reduced as an ethical matter; and (3) the intermediate three orders of magnitude, where cost-benefit balancing is common and where CBA may be the best way to decide when and how to declare the risk reduction a “mission accomplished.”

8.3 Which Small Costs Should be Ignored?

There was significant consensus among the 672 subjects that an annual regulatory cost of somewhere between \$10 and \$100, with a central tendency of \$25, was the dividing line between “essentially zero” and a cost that should be balanced against benefits. The interindividual variance in those judgments did not decrease significantly when the responses were converted to fractions of yearly (weekly) salary, with still about a 10-fold interquartile range (between 1/3 and 3 hours of work at the respondent’s current salary).

Although no regulatory agency to our knowledge has ever rounded down tiny individual costs to zero, ***it seems crucially important to us that this should occur if any explicit or implicit rounding down is to occur on the risk side of the ledger.*** So, just as 100 million people each facing a lifetime risk of 1/770,000 (the median of responses for *de minimis* risk) arguably should not be construed as worthy of any regulatory intervention—even though a total of about 130 “statistical lives” would be “lost” over 70 years if nothing was done—parity dictates that an intervention that imposed a \$25 annual cost upon each of those 100 million citizens should not be counted as affecting economic welfare *at all*. Even though \$2.5 billion was being spent in this example, from the point of view of a large number of “invisible” and immaterial changes in personal welfare, it is reasonable to conclude that such an intervention has no welfare effects worthy of attention. ***If we are going to continue to embrace the notion that some risks are too small to matter and hence never worth reducing, we at least need to consider that some costs may be too small to matter, as long as they are spent in service of providing non-trivial health/environmental benefits.***

The \$2.5 billion figure above is quite different from the definition that OIRA has used since the 1980s, which treats any regulation with more than a \$100 million annual impact on the economy as a “major rule” requiring additional scrutiny. Perhaps this cutoff is a relic that should be rethought. However, the real issue here is that *what is “major” should depend on effects to individuals*, and that these effects can be large or small depending on how many citizens actually share in the economic burdens of a regulation. A \$90 million rule that imposes costs of \$1000 on each of 90,000 persons is clearly more burdensome, perhaps infinitely more so, than a \$100 million rule whose cost is split 100 million ways!

8.4 Which Costs are so Large that their Imposition Should be Valued at (Far) More than the “Raw Dollars Spent”?

The finding that most subjects consider an involuntary cost of somewhere between about \$300 and \$3000 annually to be an “intolerable” amount to contribute to society for a given public-health intervention (with a central tendency of roughly \$800) suggests that while few if any regulations will exceed the multi-billion-dollar *total* cost that would result from everyone in the

nation being subjected to nearly-intolerable costs, it is quite possible that many individuals will see intolerable costs *to themselves* from more modest regulations whose costs are concentrated.

The scale of this set of responses (roughly a few days' to a few weeks' salary) strongly suggests that *any* worker who loses their job, even for a brief period before being hired elsewhere, has experienced a welfare loss that ought not to be treated merely as a tiny portion of some total-dollar-cost amount that is "worth society's spending" for the health benefits produced. A recent compilation of scholarship about the likely unemployment consequences of environmental/health/safety regulation (Coglianese, Finkel, and Carrigan, 2014) concluded that these rules rarely cause significant amounts of unemployment, but that the small pockets of economic harm in their wake deserve more attention. As with the discussion of intolerable risks above, we urge scholars to begin to develop principled ways to convert economic harms into units of welfare rather than treating "every dollar as a dollar" regardless of its magnitude to the individual, and our forthcoming article reporting on elicitations of perceived disutility as individual cost increases may help begin such an effort.

It is essential to mention, though, that since many regulations that cause some unemployment also spur new hiring in other sectors of the economy, there may well be substantial economic benefits on the "cost side" of a regulation that should also be expressed in a welfare metric. *Net* job losses may introduce some intolerable costs, but net job gains do the opposite.

8.5 Comparing Risk and Cost Benchmarks Sheds Light on Cost-Benefit Balancing.

The entire foundation of benefit-cost analysis for environmental/health regulation rests on the assumption that laypeople's stated or revealed answers to questions of the form "what amount of dollars is equivalent to this amount of risk?" (averaged over a suitable sample) allow decision-makers to convert risk reduction benefits to monetary units. Therefore, we believe it is reasonable to "line up" the average responses for *de minimis* risk and cost, and for intolerable risk and cost, and ask what "value of life" these two mathematical identities imply. Indeed, we suggest that equating a pair of well-defined benchmarks (the amount of risk or cost that divides

trivial from non-trivial, or tolerable from intolerable) is at least as reasonable, perhaps more so, than asking laypeople to trade off arbitrary amounts of risk that do *not* have particular salience for their equivalent cost. This exercise results in two new estimates of the value of a statistical life (VSL):

- On average, respondents in this study found that an excess risk of 1 in 770,000 per lifetime was *de minimis*, and also found that a cost of \$25/year was *de minimis*. Since at a 3% discount rate, a stream of \$25 annual costs has a present value of about \$725 over 70 years, these two results imply that \$725 and 1/770,000 are equivalent, which in turn implies a VSL of **\$558 million**, about 50 times higher than the canonical value used by regulators today. This comparison should not be sensitive, as the VSL surely is, to a respondent's ability to pay, as it merely asks for a pair of risk and cost values that are similarly situated at the lower boundary of salience.

When researchers have asked laypeople how much they are willing to pay to eliminate a risk of 1/million, the average answer tends to be around \$10, implying a VSL of about \$10 million. We did not directly ask subjects how much they were willing to pay to avert a personal mortality risk of 1/770,000 (instead, several of our recent papers (Johnson and Finkel, 2016; Finkel and Johnson, 2018; Johnson and Finkel, 2022) recommend a “macro-risk” approach, which explicitly asks subjects to trade off a given number of lives prolonged across the nation for the cost of the life-prolonging intervention). Nevertheless, our subjects have responded that a payment of \$25 per year is no more salient than the \$10 lifetime payment deemed elsewhere to be acceptable to avert a 1/million risk. ***This disparity suggests that stated-preference elicitations of the VSL might greatly underestimate its true value, because laypeople don't consider such a small risk, or the small monetary bids they are typically asked to accept or reject to avert it, to be of meaningful scale to them.***

- At the other end of the spectrum, respondents on average regarded a lifetime risk of 1/570 or greater as intolerable, while at the same time they regarded an annual cost of \$940 or greater as intolerable. Converting a stream of annual costs into a 70-year cost (\$28,450/lifetime) results in an estimate of the VSL of **\$16.2 million**, which is close to (but somewhat higher than) the conventional \$10 million figure.

8.6 Possible Changes to Risk Management in Light of These Findings.

The results presented here amount, most simply, to a repudiation of the idea that any level of risk or of cost can or should be converted to units of welfare via some scalar conversion factor. Instead, we conclude that any individual risk or individual cost must first be categorized as belonging to one of three distinct regions: smaller than *de minimis*, greater than intolerable, or some amount in between.

This insight immediately suggests that no risk or cost affecting the public can adequately be summarized, evaluated, or responded to without considering the distribution of individual welfare effects: *considering only the raw total of dollars expended or lives prolonged simply misses much of the real-world impact, and can lead to errors of gross over-regulation or tragic under-protection.* A large number of diffuse and trivial harms are *not* as worthy of attention as their raw totals might imply, and a smaller number of concentrated and dire harms should not be discounted because they don't result in "enough" total cost or "enough lives saved."

We have seen in the literature and commentary over the decades a few glimpses of the realization that something is amiss. For no other reason than that this pair of articles (written 13 years apart) have the same first author, and one who has held the rarefied position as the White House's "regulatory czar," we offer two quotes from Cass Sunstein:

- "Fearsome risks—those that stimulate strong emotional responses, such as fear and anxiety—usually... tend to have extremely low probabilities, since life today is no longer nasty, brutish and short... In the face of a fearsome risk, people often exaggerate the benefits of preventive, risk-reducing, or ameliorative measures. In both personal life and politics, the result is damaging overreactions to risks." (Sunstein and Zeckhauser, 2008).
- "Suppose that a \$900 million [regulatory] expenditure will be spread across at least 200 million people... Most people will lose essentially no welfare from an annual \$4 loss..." (and he concludes that it would be folly to *fail* to control a hazard at this \$900 million price if 100 lives could be saved in the doing) (Sunstein 2021).

Together, these insights offer a cautionary note to decision-makers who might give too much deference to widely-dispersed but always trivial harms, whether on the risk side or the cost side of the ledger. Experts in our field rarely use even-handed language to decry “damaging overreactions to costs” (Finkel 2021), but the principle stands that we should not let the number (or the “volume,” in summed-up decibel units!) of persons facing essentially non-existent harms to dictate public policies.

And we hope it is obvious by now that the “environmental justice” movement, as well as the special attention so often given to the economic impacts on small business and to the possible unemployment effects of regulation, are fundamentally motivated by the concern that intolerable effects on subpopulations are shortchanged if we only lump them into the raw totals of dollars spent and lives saved.

Combining the concerns about over-reaction to trivial effects and under-reaction to intolerable effects leads directly to these four recommendations about four paradigm situations where the conventional total-costs-versus-total-benefits comparison may fail:

- (1) Where risks (affecting portions of society) are far above *de minimis* levels, but where the vast majority of the population faces *de minimis* costs. Here we might rationally intervene to **control** a hazard even though “ $C > B$ ” as measured simplistically. For example, traditional CBA would reject a policy intervention that eliminated a 1/100 risk to 1000 persons (10 “statistical lives”), if 100 million people each had to pay \$1 to fund the intervention (a \$100 million cost for a \$90 million benefit at $VSL = \$9$ million). But in this example 1000 citizens would be freed from intolerable risk, and *no one* would face a meaningful cost.
- (2) Where costs to some are far above *de minimis* levels, but the entire population faces largely *de minimis* risks. Here we might rationally **ignore** a hazard even though “ $B > C$ ” as measured simplistically. For example, traditional CBA would favor an intervention that eliminated a risk of 1.5×10^{-7} to each of 100 million people (15 “statistical lives,” or a monetized benefit of \$135 million), even if the intervention

made 100 workers permanently unemployable, at a cost to each of them of \$1 million in lifetime present value. But in this example 100 citizens would face intolerable cost, for the sake of reducing trivially small risks.

- (3) Where costs to some are above *de minimis* amounts, but risks to some are above intolerable levels, we might rationally **control** a hazard even though “ $C > B$ ” as measured simplistically (here, of course, we might partially reduce only the highest risks).
- (4) Where risks to some are significant, but costs to some are above intolerable levels, we might rationally **ignore** a hazard even though “ $B > C$ ” as measured simplistically (here, of course, we might eschew those aspects of the intervention that impose intolerable costs, and partially reduce the remaining risks).

For shorthand, one might divide these four cases into “two to pursue, two to eschew”—as in “pursue low-regret protection” (#1 above) and “pursue worthwhile justice” (#3), but also “eschew expensive perfection” (#2) and “eschew ruinous amelioration” (#4). These are four responses to limiting-case situations that could correct some common mistakes in risk management, but they lack a quantitative underpinning. To conclude this article, we sketch out two quantitative decision rules via which government could de-emphasize *de minimis* risks and costs, while giving more appropriate additional weight to intolerable risks and costs.

Option 1:

Government could construe that the goal of environment/health/safety regulation is to *relieve as many citizens as possible of intolerable risk, while creating few if any intolerable costs in so doing*. If we are willing to treat intolerable risks and costs as vastly more important than any smaller harms, because these adverse effects are *sui generis* and create moral imperatives, then one simple but profound change in decision-making would be to replace the [total benefits exceed total costs] test with a test that ***requires the number of citizens relieved of intolerable risk to exceed the number subjected to intolerable costs from the intervention***. A regulation or other policy that could relieve N citizens from facing a risk greater than R_I , while imposing costs

greater than C_1 on fewer than N citizens, could be regarded as “doing more good than harm,” albeit in a different way than a $B > C$ criterion would define that “net plus.”

Admittedly, this test ignores harms below intolerable levels, and changes within the large region between *de minimis* and intolerable levels can be the root of much of the life-saving, or the total cost, in public policy. But the traditional [B-C] test metric ignores the things that matter most to people, which suggests that this “net reduction of intolerable harm” could be a useful *complementary* test to the standard one.

We also note that this focus on the right-hand tail of welfare effects is motivated similarly to the growing concern about “environmental justice.” However, our test does *not* require identifying subpopulations who merit more attention by virtue of characteristics unrelated to their risk or cost, such as race or being a small business owner. It only looks at the injustice of *anyone* subjected to intolerably large harms. *So instead of “effects to people that matter,” this criterion cares about “effects that matter to people.”* Needless to say, it would be possible to *additionally* give more weight to those intolerable risks or costs that also affect subpopulations deemed especially important.

In addition, we emphasize that no less an authority than the U.S. Supreme Court has made room for exactly this kind of cost-benefit thinking as a complement to [C-B] calculations. In 1980 and 1981, the Court issued two opinions about OSHA regulation that, read together, actually force OSHA to consider both cost and benefit, but in a way much more concordant with our Option 1 above than with the traditional test.

In the 1980 “Benzene” decision (Industrial Union Department, 1980), the Court ruled that OSHA must regulate risks that are large enough to be “plainly unacceptable” and cannot reduce risks below *de minimis* levels (as discussed above, the Court suggested that a lifetime excess risk of death of 1/1000 above was unacceptable, and that a risk of 1/billion or below was clearly immaterial). For a short time, this strong focus on intolerable risks lacked a countervailing decision-rule for costs—but in the 1981 “Cotton Dust” decision (American Textile Mfgs. Inst., 1981), the Court promptly clarified that OSHA could not impose costs high enough, even if in order to reduce intolerable risks, such that these costs would threaten the “fundamental

competitive structure” of an entire industry. Although no OSHA standard has ever come close to this cost constraint, the Court made clear that marginal firms within a regulated industry could be hit hard or even forced to close down, but forbade OSHA to impose costs high enough to prevent the affected workers to be re-hired by one of the firms who could more easily comply.

While OSHA sometimes likes to say that these decisions free the Agency from having to conduct CBA, the reality is that they free OSHA from having to balance total costs and total benefit algebraically, and instead require the Agency to *reduce intolerable risks without imposing intolerable costs*. This is still a benefit-cost calculus—that a policy “does good” by erasing each intolerable risk, but can do harm by imposing new intolerable costs—and this is exactly the decision-rule we offer here as a complement.

Option 2:

Supplementing a $[B-C > 0]$ test with a “net reduction in intolerable harms” test is really the limiting case of a more general refinement to the simple net benefit test. $[B-C]$ is in fact a major simplification of the logically and mathematically *correct* way to aggregate the consequences of individual risks and individual costs, thus:

$$NB = \sum_i [f_1(R_i) - f_2(C_i)]$$

where R_i and C_i are the risk and the cost to each individual, and f_1 and f_2 are different functions that map any level of risk or cost to the disutility or harm that level presents. The equation above reduces to $[B-C]$ only if f_1 is a constant scalar multiplier and f_2 is the identity function: in other words, if the monetized risk is just a constant VSL divided by the risk (e.g., \$9 for a 1/million risk, \$900,000 for a 1-in-10 risk), and if the correct way to measure the disutility of cost is to express it in its raw dollar units. Then, total benefit is just the number of statistical lives “saved” (multiplied by the VSL), and total cost is just total dollars required to fund the intervention.

This article, and much prior scholarship, makes the claim that neither simplification is warranted, and that in fact personal risk and personal cost *must* be converted to welfare units using a non-linear function for each. The findings here suggest that both functions should have a threshold, such that the harm done below *de minimis* levels is zero, and should be convex (second derivative > 0), such that the harm done above intolerable levels is vastly larger than at any smaller level.

As part of this pair of experiments, we posed structured questions so that respondents could offer their estimates of how their aversion to rising risks or costs changed with their magnitude. Because the methods here, and the results obtained, require considerable explanation, we defer sketching out the particulars of f_1 and f_2 to a subsequent article. Here we merely point out that having elicited laypersons' views about the extremes of f_1 and f_2 , the next step is to parameterize each function so that ultimately, any intervention that reduces each individual's risk and that imposes costs that can be estimated at the individual level, can be subjected to a "full individual-welfare test." The goal of such a criterion would be to show whether the intervention increases *net welfare*, rather than increasing the over-simplified concept we have come to know as "net benefit" and have come to accept as the end-all without realizing that net benefit is a special and arguably unrealistic notion.²⁸

9. Conclusions:

We hope that others will enlist a larger and nationally representative sample, and reproduce or change our four key findings: (1) that laypeople regard lifetime excess mortality risks below about 1/800,000 as unworthy of any regulatory attention; (2) they regard risks above about 1/600 as intolerable and that should never persist; (3) they regard costs of about \$25/year

²⁸ As this article was in final preparation post-review, an interesting controversy began to influence the future of regulatory cost-benefit analysis. The White House Office of Information and Regulatory Affairs proposed sweeping revisions to Circular A-4, the guidance governing CBA review in the federal system, and OIRA's enthusiasm for agencies to apply "distributional weights" to costs affecting poor citizens versus rich ones dominated the comments received (Xie, Hay, and Hirsch 2023). Some commenters (Sullivan 2023) argued that these weights are inappropriate and policy-laden, despite OIRA's clear (and we believe, correct) argument in the revisions that agencies have *always* been applying weights, merely implicit and equal ones. But others (see, e.g., Adler 2023; Finkel 2023) explicitly argued that such weights are the only ways to correct the long-standing *error* in CBA that fails to transform units of dollar cost into units of welfare decrement, as the welfare toll of a given dollar cost is clearly larger for a poor person than a rich one.

(\$725/lifetime) as *de minimis* and hence always bearable for any worthy risk-reduction purpose; and (4) they regard annual costs of about \$1000 (about 1 week's salary on average) as intolerable, hence as costs that government should be very wary of imposing on anyone in the name of risk reduction. IF, in either of the first two cases, the risks or costs were so widely distributed that every American was subject to them, our results would imply that as many as 375 statistical fatalities per 70 years could be safely ignored, or that a regulation costing as much as \$7.5 billion per year could be borne without any tangible hardship on any individual. Obviously, very few rules affect so many people that hundreds of millions of *de minimis* costs or risks could be deemed immaterial.

Whatever the estimates (or uncertainty distributions of estimates) of the four goalposts turn out to be, we think it will greatly improve environmental/health/safety policy if we estimate them empirically rather than by invoking trite rules of thumb about them. More importantly, it is time—arguably, long past time—to stop paying lip service to the concept that some effects need to be shelved while other effects need to be given much more prominence, and start adapting our risk management policies to *truly focus on intolerable harms and de-emphasize trivial ones*.

In sum, some environmental/health interventions impose intolerable costs while only reducing diffuse and trivially small risks. These result in outcomes that are inefficient, unjust, and thus unethical. Equally or more importantly, other *failures* to intervene (or half-hearted, weak interventions) fail to reduce intolerable risks, for the sake of avoiding trivial and eminently manageable costs. These perpetuate outcomes that are inefficient, unjust, and thus unethical.

Only through more research along the lines we have initiated here can science, economics, and society hope to know which policies are which.

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